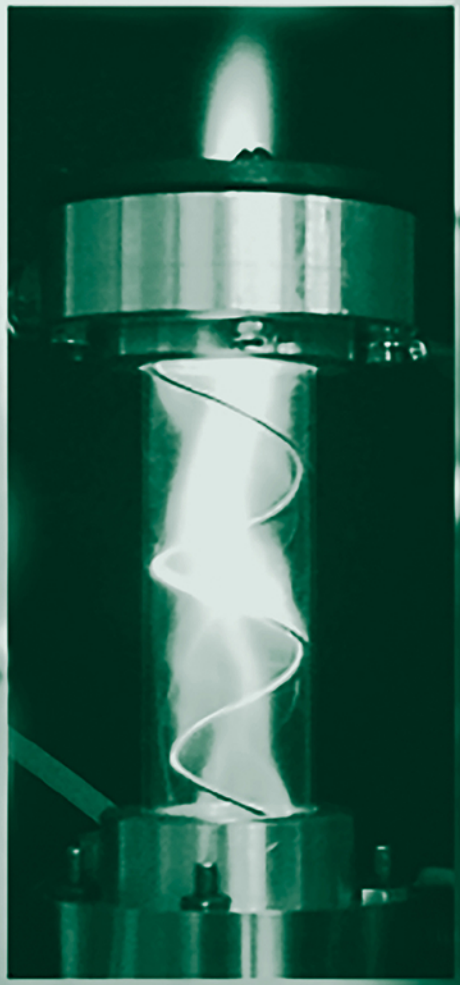
# Plasma Physics and Engineering

third edition

Alexander Fridman
Lawrence A. Kennedy



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Alexander Fridman
and Lawrence A. Kennedy

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# Preface

Plasma enjoys an important role in a wide variety of industrial processes including material processing, environmental control, electronic chip manufacturing, light sources, green energy, fuel conversion and hydrogen production, bio-medicine, flow control, catalysis, and space propulsion. It is also central to understanding most of the universe outside of earth. As such, the focus of this monograph is to provide a thorough introduction to the subject and to be a valued reference that serves engineers and scientists as well as students. Plasma is not an elementary subject and the reader is expected to have the normal engineering background in thermodynamics, chemistry, physics and fluid mechanics upon which to build an understanding of this subject. The text is organized in two parts. Part I addresses the basic physics of plasma, part II addresses the physics and engineering of electric discharges. The text is adaptable to a wide range of needs. The material has been taught to graduate and senior level undergraduate students from most engineering disciplines, chemistry, and physics. For the latter it can be packaged to focus on the basic physics of plasma with only selections from discharge applications. For graduate courses a faster pace can be set covering Part I and Part II.

This third edition of *Plasma Physics and Engineering* follows the general outline of the first two editions but includes many enhancements and some totally new subjects. A totally new Chapter 13 is introduced, focused on different approaches to generation of plasma inside of different fluids. It includes discharges in bubbles and voids, formation of nano-plasma in water from nano-tip electrodes, and finally nanosecond-pulsed plasma generation in liquids without bubbles. Such liquid plasmas are discussed in the Chapter 13 not only in water and water solutions, but also in dielectric oils, and even in liquid nitrogen. Substantial changes appear in Chapter 11 focused on discharges in aerosols and dusty plasmas. Significant totally new Section 9.7 is aided to Chapter 9 to describe different aspects of physics and engineering of nanosecond and microsecond pulsed dielectric barrier discharges. Significant new material is aided in the third edition to cover new fundamental subjects regarding plasma discharges in water in relation to its cleaning from chemical and biological impurities, as well as in relation with mechanisms and kinetics of plasma-medical processes. In addition, dozens of smaller additions, including new problems and concept questions improve the depth and breadth of the coverage of many subjects.

**Alexander Fridman**
**Lawrence A. Kennedy**


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# Part 1

## Fundamentals of Plasma Physics and Engineering


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# 1 Plasma in Nature, in the Laboratory, and in Industry

The term plasma is often referred to as the fourth state of matter. As temperature increases, molecules become more energetic and transform in the sequence: solid, liquid, gas, and plasma. In the latter stages, molecules in the gas dissociate to form a gas of atoms and then a gas of freely moving charged particles, electrons, and positive ions. This state is called the plasma state, a term attributed to Langmuir to describe the region of a discharge that was not influenced by the walls and/or electrodes. It is characterized by a mixture of electrons, ions, and neutral particles moving in random directions that on the average is electrically neutral ($n_e \cong n_i$). In addition, plasmas are electrically conducting due to the presence of these free charge carriers and can attain electrical conductivities larger than metal such as gold and copper. Plasma is not only the most energetic state of matter but also the most challenging for scientists. The temperature of the charged particles in a plasma can be so high that their collisions can result in thermonuclear reactions! Plasmas occur naturally, but also can be man-made. Plasma generation and stabilization in a laboratory and in industrial devices are not easy, but very promising for many modern applications, including thermonuclear synthesis, electronics, lasers, and many others. Most of your computer hardware are made based on plasma technologies and do not forget about the very large and thin TV plasma screens so popular today.

Plasmas offer two main characteristics for practical applications. First, they can have temperatures and energy densities greater than those produced by ordinary chemical means. Second, plasmas are able to produce, even at low temperatures, energetic species that can initiate chemicals reactions that are difficult or impossible to obtain using ordinary chemical mechanisms. The energetic species generated cover a wide spectrum of such species, for example, charged particles including electrons, ions, and radicals, highly reactive neutral species such as reactive atoms (e.g., O, F, etc.), excited atomic states, reactive molecular fragments, and different wavelength photons. Plasmas can also initiate physical changes in material surfaces. Applications of plasma can provide major benefits over existing methods. Often processes can be performed that are not possible in any other manner. Plasma can provide an efficiency increase in processing methods and often can reduce the environmental impact when compared to more conventional processes.

## 1.1 OCCURRENCE OF PLASMA, NATURAL AND MAN-MADE PLASMAS

Although somewhat rare on Earth, plasmas occur naturally and comprise the majority of the universe encompassing among other phenomena, the solar corona, solar wind, nebula, and the earth's ionosphere. In the earth's atmosphere, plasma is often observed as a transient event in the phenomenon of lightning strokes. Since air is normally nonconducting, large potential differences can be

generated between clouds and earth during storms. Lightning discharges occur to neutralize the accumulated charge in the clouds. Such dischargers occur in two phases. First, an initial leader stroke progresses in steps across the potential gap between clouds or the earth. This leader stroke creates a low degree of ionization in the path and provides the conditions for the second phase, the return stroke to occur. The return stroke creates a highly conducting plasma path for the large currents to flow and neutralize the charge accumulation in the clouds. At altitudes of approximately 100 km, the atmosphere no longer remains nonconducting due to its interaction with solar radiation. From the energy absorbed through these solar radiation processes, a significant number of molecules and atoms become ionized and make this region of the atmosphere a plasma. As one progresses further into near-space altitudes, the earth's magnetic field interacts with charged particles streaming from the sun. These particles are diverted and often become trapped by the earth's magnetic field. The trapped particles are most dense near the poles and account for the Aurora Borealis. Lightning and the Aurora Borealis are the most common natural plasma observed on earth and one may think that these are exceptions. However, the earth's atmospheric environment in which we live is an anomaly; we live in a bubble of unionized gas surrounded by plasma. Lightning occurs at relatively high pressures and the Aurora Borealis at low pressures. Pressure has a strong influence on plasma, influencing its respective luminosity, energy or temperature, and the plasma state of its components. Lightning consists of a narrow, highly luminous channel with numerous branches. The Aurora Borealis is a low luminosity, diffuse event. Since plasmas occur over a wide range of pressures, it is customary to classify them in terms of electron temperatures and electron densities. Figure 1.1 provides a representation of the electron temperature (in electron volts, eV) and electron densities (in $1/cm^3$) typical of natural and man-made plasmas. The electron temperature is expressed in electron volts (eV); one electron-volt is equal to approximately 11,600 K. Man-made plasma ranges from slightly above room temperature to temperatures comparable to the interior of stars. Electron densities span over 15 orders of magnitudes. However, most plasmas of practical significance have electron temperatures of 1–20 eV with electron densities in the range $10^6$–$10^{18}$ $1/cm^3$. Not all particles need to be ionized in a plasma; a common condition is for the gases to be partially

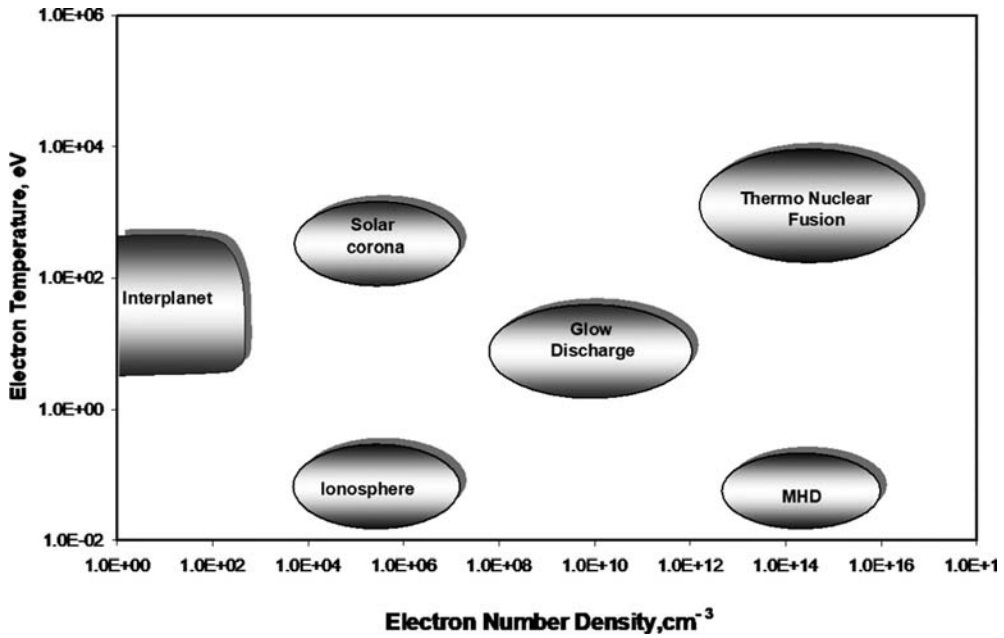



**FIGURE 1.1** Operating regions of natural and man made plasma.

ionized. Under the latter conditions, one must examine the particle densities to determine if it is plasma. If there is a very low density of charged particles, the influence of the neutral particles can swamp the effects of interaction between charged ones. Under these conditions, the neutral particles dominate the collision process. The force laws appropriate to the specific particles govern collisions. For neutral particles, collisions are governed by forces that are significant only when the neutrals are in close proximity to each other. Conversely, the longer-range Coulomb Law governs collisions between charged particles. Hence, the behavior of the collision will change as the degree of ionization increases with the Coulomb interactions becoming more important. Coulomb's Law will govern all particles in the fully ionized plasma.

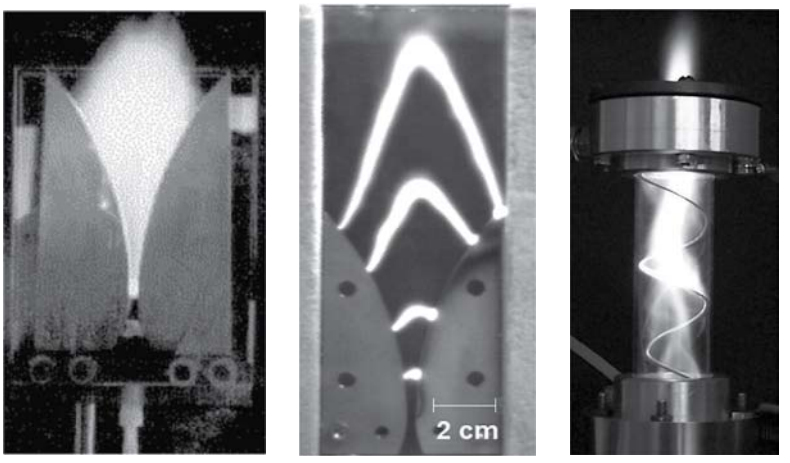
Langmuir was one of the early pioneers who studied gas discharges and defined plasma to be a region not influenced by its boundaries. The transition zone between the plasma and its boundaries was termed the plasma sheath. The properties of the sheath differ from those of the plasma, and these boundaries influence the motion of the charged particles in this sheath. They form an electrical screen for the plasma from the influences of the boundary. As in any gas, the temperature is defined by the average kinetic energy of the particle, molecule, atom, neutral, or charge. Thus, plasmas will typically exhibit multiple temperatures unless there are sufficient collisions between particles to equilibrate. However, since the mass of an electron is much less than the mass of a heavy particle, many collisions are required for this to occur.

The use of electric discharge is one of the most common ways to create and maintain a plasma. Here, the energy from the electric field is accumulated by the electron between collisions that subsequently transfers a portion of this energy to the heavy neutral particles through collisions. Even with a high collision frequency, the electron temperature and heavy particle temperature normally will be different. Since the collision frequency is pressure-dependent, high pressures will increase the collision frequency and the electron's mean free path between collisions will decrease. One can show that the temperature difference between electrons and heavy neutral particles is proportional to the square of the ratio of the energy an electron receives from the electric field (**E**) to the pressure ($p$). Only in the case of small values of $\mathbf{E}/p$, does the temperatures of electrons and heavy particles approach each other. Thus, this is a basic requirement for *Local Thermodynamic Equilibrium* (LTE) in a plasma. Additionally, LTE conditions require chemical equilibrium as well as restrictions on the gradients. When these conditions are met, the plasma is termed a *thermal plasma*. Conversely, when there are large departures from these conditions, $T_e > T_n$, the plasma is termed a *nonequilibrium plasma* or *nonthermal plasma*. As an example, for plasma generated by glow discharges, the operating pressures are normally less than 1 kPa and have electron temperatures of the order of $10^4$ K with ions and neutral temperatures approaching room temperatures. Physics, engineering aspects, and application areas are quite different for thermal and nonthermal plasmas. Thermal plasmas are usually more powerful, while nonthermal plasmas are more selective. However, these two so different types of ionized gases have many more features in common and both are plasmas.

## 1.2 GAS DISCHARGES

There are many well-established uses of plasma generated by an electric discharge. Historically, it was the study of such discharges that laid the initial foundation of much of our understanding of plasma. In principle, discharges can be simply viewed as two electrodes inserted into a glass tube and connected to a power supply. The tube can be filled with various gases or evacuated. As the voltage applied across the two electrodes increases, the current suddenly increases sharply at a certain voltage characteristic of an electron avalanche. If the pressure is low, of the order of a few Torr, and the external circuit has a large resistance to prohibit a large current flow, a glow discharge develops. This is a low-current, high-voltage device in which the gas is weakly ionized (plasma). Glow discharges are widely used and are a very important type of discharge. They form the basis of fluorescent lighting. Such a discharge is present in every fluorescent lamp and the voltage–current characteristics of such discharges are widely employed in constant voltage gas tubes used in

electronic circuits. Other common gas vapor discharges used for lighting are mercury vapor and neon. In power electronics applications, the need for switching and rectification has been addressed with the development of the thyratron and ignitron, which are important constant voltage gas tubes. If the pressure is high, of the order of an atmosphere, and the external circuit resistance is low, a thermal arc discharge can form the following breakdown. Thermal arcs usually carry large currents, greater than 1 A at voltages of the order of tens of volts. Further, they release large amounts of thermal energy. These types of arcs are often coupled with a gas flow from high-temperature plasma jets. A corona discharge occurs only in regions of sharply nonuniform electric fields. The field near one or both electrodes must be stronger than in the rest of the gas. This occurs near sharp points, edges, or small diameter wires. These tend to be low power devices limited by the onset of the electrical breakdown of the gas. However, it is possible to circumvent this restriction through the use of pulsating power supplies. Many other types of discharges are variations or combinations of these. The gliding arc is one such example. Beginning as a thermal arc located at the shortest distance between the electrodes, it moves with the gas flow at a velocity about 10 m/s and the length of the arc column increases together with the voltage. When the length of the gliding arc exceeds its critical value, heat losses from the plasma column begins to exceed the energy supplied by the source, and it is not possible to sustain the plasma in a state of thermodynamic equilibrium. As a result, a fast transition into a nonequilibrium phase occurs. The discharge plasma cools rapidly to a gas temperature of about $T_n$ = 1,000 K and the plasma conductivity is maintained by a high value of the electron temperature $T_e$ = 1 eV (about 11,000 K). After this fast transition, the gliding arc continues its evolution, but under nonequilibrium conditions ($T_e >> T_n$). The specific heat losses in this regime are much smaller than in the equilibrium regime (numerically about three times less). The discharge length increases up to a new critical value approximately three times its original critical length. The main part of the gliding arc power (up to 75%–80%) can be dissipated in the nonequilibrium zone. After the decay of the nonequilibrium discharge, the evolution repeats from the initial break down. This permits the stimulation of chemical reactions in regimes quite different from conventional combustion and environmental situations. It provides an alternate approach to addressing energy conservation and environmental control. Figure 1.2 illustrates the gliding arc. Many of the features in these discharges are also typical for discharges in rapidly oscillating fields where electrodes are not required. As such, the classification of discharges has been developed, which does not involve electrode attributes. Such a classification is based upon two properties: (1) the state of the ionized gas and (2) the




(a) (b) (c)

**FIGURE 1.2** (a) Gliding arc viewed with normal photography, (b) Gliding arc viewed with high-speed photography, (c) Vortex gliding arc.

**TABLE 1.1**
**Classification of Discharges**

| | Breakdown | Nonequilibrium Plasma | Equilibrium Plasma |
|---|---|---|---|
| Constant E Field | Initiation of glow discharge | Glow discharge | High-pressure arc |
| Radio frequencies | Initiation of RF discharge in rarefied gas | Capacitively coupled RF discharge in rarefied gas | Inductively coupled plasma |
| Microwave | Breakdown in waveguides | Microwave discharge in rarefied gas | Microwave plasmatron |
| Optical range | Gas breakdown by laser radiation | Final stage optical breakdown | Continuous optical discharge |

frequency range of the electric field. The state of the ionized gas distinguishes between (1) breakdown of the gas, (2) sustaining a nonequilibrium plasma by the electric field, and (3) sustaining an equilibrium plasma. The frequency criteria classify the field into (1) DC, low frequency, and pulsed E fields; (2) radio frequency E fields, (3) microwave fields, and (4) optical fields. These are summarized in Table 1.1.

## 1.3 PLASMA APPLICATIONS, PLASMAS IN INDUSTRY

Although this text is mainly focused on the fundamentals of plasma physics and engineering, ultimately we are interested in the application of plasmas; see the following book Fridman "Plasma chemistry" (2008) focused on the applications. The number of industrial applications of plasma technologies is extensive and involves many industries, especially electronics, lighting, coatings, treatment and processing of materials, metallurgy, and energy systems. High-energy efficiency, specific productivity, and selectivity may be achieved in plasmas for a wide range of chemical processes. As an example, for $CO_2$ dissociation in nonequilibrium plasmas under supersonic flow conditions, it is possible to selectively introduce up to 90% of total discharge power in CO production when the vibrational temperature is about 4,000 K and the translational temperature is only 0 (300 K). The specific productivity of such a supersonic reactor achieves 1,000,000 L/h, with power levels up to 1 MW. The key point for practical use of any chemical process in a particular plasma system is to find the proper regime and optimal plasma parameters among the numerous possibilities intrinsic to systems far from equilibrium. In particular, it is desired to provide high operating power for the plasma chemical reactor together with high selectivity of the energy input while simultaneously maintaining nonequilibrium plasma conditions. Generally, two very different kinds of plasmas are used for applications. Thermal plasma generators have been designed for many diverse industrial applications covering a wide range of operating power levels from less than 1 kW to over 50 MW. However, in spite of providing sufficient power levels, these are not well adapted to the purposes of plasma chemistry, where selective treatment of reactants (through the excitation of molecular vibrations or electron excitation) and high efficiency are required. The main drawback of using thermal plasmas is overheating the reaction medium when energy is uniformly consumed by the reagents into all degrees of freedom and hence, high energy consumption required to provide special quenching of the reagents, and so on. Because of that, the energy efficiency and selectivity of such systems are rather small. Promising plasma parameters are achievable in microwave discharges. The skin effect here permits simultaneous achievement of a high level of electron density and a high electric field (and hence a high electronic temperature as well) in the relatively cold gas. Existing super high frequency discharge technology can be used to generate dense ($n_e = 10^{13}$ electrons/cm$^3$) nonequilibrium plasmas ($T_e$ = 1–2 eV, $T_V$ = 3,000–5,000 K, $T_n$ = 800–1,500 K, for supersonic flow $T_n \leq 150$ K and less) at pressures up to 200–300 Torr and at power levels reaching 1 MW.

An alternative approach for plasma-chemical gas processing is the nonthermal one. Silent discharges such as glow, corona, short pulse, microwave or radio-frequency (RF) electrical discharges are directly produced in the processed gas, mostly under low pressure. The glow discharge in a low-pressure gas is a simple and inexpensive way to achieve a nonthermal plasma. Here, the ionization processes induced by the electric field dominate the thermal ones and give relatively high energy electrons as well as excited ions, atoms, and molecules which promote selective chemical transitions. However, the power of glow discharges is limited by the glow to arc transition. Gas, initially below 1,000 K, becomes hot (>6,000 K), and the electron temperature, initially sufficiently high (>12,000 K), cools close to the bulk gas temperature. The discharge voltage decreases during such a transition making it necessary to increase the current in order to keep the power on the same level which in turn leads to thermalization of the gas. Thus, cold nonequilibrium plasmas created by conventional glow discharges offer good selectivity and efficiency, but at limited pressure and power levels. In these two general types of plasma discharges, it is impossible to simultaneously keep a high level of nonequilibrium, high electron temperature, and high electron density, whereas most perspective plasma chemical applications simultaneously require high power for high reactor productivity and a high degree of nonequilibrium to support selective chemical process. Recently, a simpler technique offering similar advantages has been proposed, the GlidingArc. Such a gliding arc occurs when the plasma is generated between two or more diverging electrodes placed in a fast gas flow. It operates at atmospheric pressure or higher and the dissipated power at nonequilibrium conditions reaches up to 40 kW per electrode pair. The incontestable advantage of the GlidingArc compared with microwave systems is its cost; it is much less expensive compared to microwave plasma.

## 1.4 PLASMA APPLICATIONS FOR ENVIRONMENTAL CONTROL

Low-temperature, nonequilibrium plasmas are an emerging technology for abating low VOC emissions. These nonequilibrium plasma processes have been shown to be effective in treating a wide range of emissions including aliphatic hydrocarbons, chlorofluorocarbons, methyl cyanide, phosgene, formaldehyde, as well as sulfur and organophosphorus compounds. Such plasmas may be produced by a variety of electrical discharges or electron beams. The basic feature of nonequilibrium plasma technologies is that they produce plasma in which the majority of the electric energy (more than 99%) goes into the production of energetic electrons instead of heating the entire gas stream. These energetic electrons produce excited species, free radicals, and ions as well as additional electrons through the electron impact dissociation, excitation and ionization of the background molecules. These excited species, in turn, oxidize, reduce, or decompose the pollutant molecules. This is in contrast to the mechanism involved in thermal processes that require heating the entire gas stream in order to destroy pollutants. In addition, the low-temperature plasma technology is highly selective and has relatively low maintenance requirements. Its high selectivity results in relatively low energy costs for emissions control, while low maintenance keeps annual operating expenses low. Furthermore, these plasma discharges normally are very uniform and homogeneous, which results in high process productivity. Although the products of these plasma processes are virtually indistinguishable from incineration products ($CO_2$, $H_2O$, $SO_2$, etc.), chemical reactions occurring in these technologies are substantially different than in incineration. Large electric fields in discharge reactors create conditions for electric breakdowns during which many electron–ion pairs are formed. Primary electrons are accelerated by the electric field and produce secondary ionization, and so forth. Created excited species, atoms, radicals, molecular and atomic ions, electrons, and radicals are capable of interacting to a certain degree with VOCs. Table 1.2 illustrates different plasma approaches to VOC destruction. Electron beams produce a plasma of the best quality with an average electron energy of 5–6 eV. Pulsed corona discharges stand very close (3–5 eV). But corona discharges have the advantage of lower capital cost. Generally, both technologies have already found applications for gas treatment form $NO_x$ and $SO_2$ emissions in coal power plants. The capability of treating large gas streams makes e-beams and pulsed corona discharges the primary candidates

**TABLE 1.2**
**Comparison of Nonthermal Plasma Technologies**

| Plasma Technology | Average Electron Energy | Capital Cost, $/W | Unit Capacity, kW | Energy Consumption, kWh/m$^3$ | Known Applications | Technical Limitations |
|---|---|---|---|---|---|---|
| Electron beam technology | 5–6 eV | 2.0 | 1,000 | 10–70 | Removal of $SO_2$, NOx in thermal power plants | Initial energy of electrons in the beam is limited by 1.5 MeV due to X-ray production |
| Gas-Phase Corona Reactor (GPCR) | 2–3 eV | 2.1 | 20 | 6–70 | Electrostatic Precipitators | The voltage cannot be as high as in pulsed corona due to spark formation; packed bed limits flow rates. |
| Pulsed Corona Discharge | 3–5 eV | 1.0 | 10 | 20–150 | Removal of polyatomic hydrocarbons in aluminum plant | Voltage is limited by the electrical network coupling and capacitance and inductance of the plasma reactor |
| Dielectric-Barrier Discharge (DBD) | 2–3 eV | 0.2 | 0.1 | 20–200 | Removal of $SO_2$, NOx, industrial ozone generators | The voltage cannot be as high as in pulsed corona due to spark formation; low unit capacity |
| Gliding Arc | 1–1.5 eV | 0.1 | 120 | 20–250 | Laboratory study on removal of methanol, o-xylene | Discharge is nonuniform, high temperatures of the gas, possible NOx generation. |

considered for VOC abatement. Here we just discussed one specific plasma application for environmental control in order to introduce some general parameters characterizing such processes.

## 1.5 PLASMA APPLICATIONS IN ENERGY CONVERSION

Plasma applications in energy conversion are many. The most publicized are those related to control thermonuclear fusion reactions, magnetohydrodynamic (MHD) power generation, and thermionic energy conversions. Fusion reactions are the source of the sun's energy. It is also the basis of our thermonuclear weapons and the more peaceful application goal of power generation. To undergo fusion reactions, the nuclei must interact very closely by overcoming the electrostatic repulsive force between particles. Thus, they must approach each other at high velocities. If one simply employs opposing beams of particles to do this, too many particles would be scattered to make this a practical approach. One must reflect the scattered particles back into the reactor as often as necessary to generate the fusion reaction. For deuterium, the required temperature exceeds 100 million degrees Kelvin to produce a significant amount of fusion. At these temperatures, any gas is fully ionized and therefore a plasma. The engineering problem for controlled fusion is then to contain this high-temperature gas for a sufficiently long time interval for the fusion reactions to occur and then extract the energy. The only way to contain such a plasma is to exploit its interactions with magnetic fields. This long-term effort is being pursued at national laboratories in many countries.

The MHD generator exploits the flow of plasma interacting with a magnetic field to generate electrical power. From Maxwell's equations, the interaction of a flowing plasma with a magnetic field ($\mathbf{B}$) will give rise to an induced electric field ($\mathbf{E} = \mathbf{v} \times \mathbf{B}$). Energy can be extracted if electrodes are placed across this flow permitting a current flow through the plasma. This in turn gives rise to a force ($\mathbf{J} \times \mathbf{B}$) opposite to the flow which decelerates it. A variation of this concept is the plasma accelerator wherein one configures the applied fields to create an accelerating force. Thermionic generators are another example of a plasma energy converter. The simplest configuration is a basic diode where the anode is cooled and a heated cathode causes electrons to boil off. The region between the electrodes is filled with cesium vapor which can be easily ionized either through contact with the hot cathode or through multiple steps in the gas. The space charge is neutralized by the positive cesium ions and allows high current to flow. Many different configurations are possible and they provide power generation for space vehicles and mobile power generators. They also have been proposed for miniature mobile power units to replace batteries. Serious attention is now directed to plasma applications for hydrogen production from different hydrocarbons, including coal, biomass, and organic wastes. Plasma discharges can be effectively combined in this case with catalytic and different hybrid systems.

## 1.6 PLASMA APPLICATION FOR MATERIAL PROCESSING

Plasma applications abound in material processing. Initially, thermal plasmas were employed due to the high temperatures they achieved over those obtainable from gaseous flames. While combustion flame temperatures are limited to approximately 3,300 K, a small plasma jet generated by the gas flow through a thermal electric arc provides a working temperature of approximately 6,500 K. When one considers the melting and/or vaporization points of many metals and ceramics, it is obvious that these arcs are extremely useful processing tools. More powerful arcs can generate temperatures to 20–30,000 K which provides environments for activated species surface modifications. The use of electric arcs for welding, melting of metals, high purity metal processing is well established. Plasma is widely employed in the coating industry where its large enthalpy content, high temperatures, and large deposition rates are advantageous for increased throughputs. It provides the ability to mix and blend materials that are otherwise incompatible. Complex alloys, elemental materials, composites, and ceramics can all be deposited. Among others, it is the preferred process for the manufacture of electronic components, wear and heat resistant coatings. With the advent of nanostructured metals,

nanopowder production and in-situ coating of these nanoparticles are accomplished with plasma processing.

Surface modification of classes of material is performed using plasma environments. For example, while the choice of a polymer for a specific application is generally driven by the bulk properties of the polymer, many common polymer surfaces are chemically inert and, therefore, pose challenges for use as substrates for applied layers. Typical applications where wetting and adhesion are critical issues are printing, painting, and metallization. Surface modification by plasma treatment of polymer surfaces can improve wetting, bonding, and adhesion characteristics without sacrificing desirable bulk properties of the polymer base material. Plasma treatments have proven useful in promoting adhesion of silver to poly(ethylene terephthalate). In addition, plasma treatments of polyester have been demonstrated to promote adhesion of gelatin containing layers such as used in the production of photographic film. In both of these examples, low-frequency capacitively coupled nitrogen plasmas have proven particularly effective.

## 1.7 BREAKTHROUGH PLASMA APPLICATIONS IN MODERN TECHNOLOGY

The most exciting applications of plasma are those that have no analogies and no (or almost no) competing processes. A good relevant example is plasma applications in microelectronics, such as etching of deep trenches in single-crystal silicon, which is so important in the fabrication of integrated circuits. Capabilities of plasma processing in microelectronics are extraordinary and unique. We probably would not have achieved such powerful and compact computers and cell phones without plasma processing. When all alternatives fail – plasma remains as a viable and valuable tool. There are other applications where plasma processes are not only highly efficient, but actually unique. For example, there are no other technologies that have competed with plasma for the production of ozone (for more than 100 years); thermonuclear plasma is a unique major future source of energy; low-temperature fuel conversion with the production of hydrogen without $CO_2$ exhaust is another example of a unique breakthrough energy technology. Another novel and unique plasma application is in water cleaning from PFOS/PFOA, the fluorinated organic compounds, deadly and extremely resistive to existing water cleaning procedures. Again in this case, it looks like plasma and is a unique effective way to do the job.

## 1.8 PLASMA BIOLOGY AND PLASMA MEDICINE

The application that shifts the paradigm in therapeutics, wound healing, disease control, cosmetics, as well as food protection from pathogens, sterilization, abatement of dangerous microorganisms, including coronavirus is in *the use of plasmas for biological and biomedical fields*. These applications, which are growing at a rapid rate, have opened new possibilities to treat diseases and wounds that are resistant to traditional methods, which includes plasma treatment of complicated ulcers and healing of cancer.


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# 2 Elementary Processes of Charged Species in Plasma

## 2.1 ELEMENTARY CHARGED PARTICLES IN PLASMA AND THEIR ELASTIC AND INELASTIC COLLISIONS

Because plasma is an ionized medium, the key process in plasma is ionization. This means the conversion of a neutral atom or molecule into a positive ion and also an electron, as illustrated in Figure 2.1a. The main two participants of the ionization process, **electrons and positive ions**, are at the same time, the most important charged particles in plasma. Normally, concentrations or number densities of the electrons and the positive ions are equal or near equal in quasi-neutral plasmas, but in "electronegative" gases, it can be different. Electronegative gases, such as $O_2$, $Cl_2$, $SF_6$, $UF_6$, and $TiCl_4$, consist of atoms or molecules with a high electron affinity (EA), which means they strongly attract electrons. Electrons stick to such molecules or collide with them with the formation of **negative ions**, the third

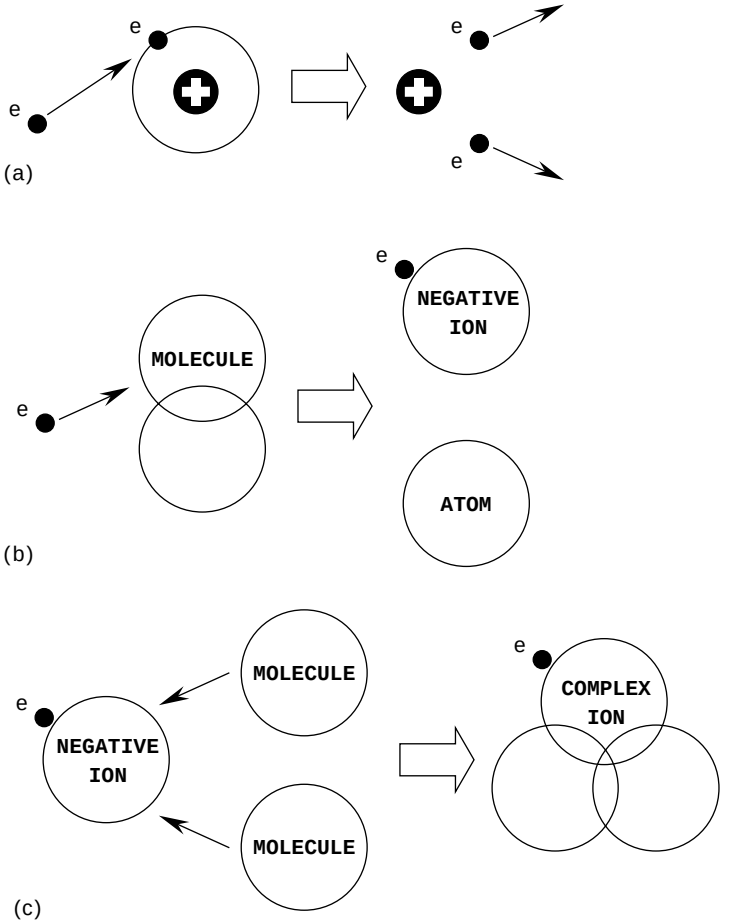



**FIGURE 2.1** Illustrations of elementary processes of: (a) ionizations, (b) dissociative attachment and formation of a negative ion, (c) complex ion formation.

important group of charged particles in plasmas. The concentration of negative ions can exceed those of electrons in electronegative gases. Illustrations of electron attachment and negative ion formation are shown in Figure 2.1b. Charged particles can also appear in plasma in more complicated forms. In high-pressure and low-temperature plasmas, the positive and negative ions attach to neutral atoms or molecules to form quite large **complex ions** or ion clusters, for example, $N_2^+N_2$ ($N_4^+$), $O^-CO_2$ ($CO_3^-$), $H^+H_2O$ ($H_3O^+$), Figure 2.1c. Ion-molecular bonds in complex ions are usually less strong than regular chemical bonds, but stronger than intermolecular bonds in neutral clusters.

### 2.1.1 Electrons

Electrons are elementary, negatively charged particles with a mass of about 3–4 orders of magnitude less than the mass of ions and neutral particles (electron charge is an elementary one: $e = 1.6\ 10^{-19}$ C, mass $m_e = 9.11\ 10^{-31}$ kg). Because of their lightness and high mobility, electrons are the first particles in receiving energy from electric fields. Afterward, electrons transmit the energy to all other plasma components, providing ionization, excitation, dissociation, etc.

Electrons are energy providers for many plasma chemical processes. The rate of such processes depends on how many electrons have sufficient energy to do the job. This can be described by means of the **electron energy distribution function** $f(\varepsilon)$, which is a probability density for an electron to have the energy ε (probability for an electron to have energy between ε and ε + Δ, divided by Δ). Quite often, this distribution function strongly depends on the electric field and gas composition and can be very far from equilibrium. Sometimes, however, (even in non-equilibrium plasmas of nonthermal discharges), the $f(\varepsilon)$ depends mostly on electron temperature $T_e$ and can then be defined by the quasi-equilibrium **Maxwell–Boltzmann distribution function:**

$$f\left(\varepsilon\right) = 2\sqrt{\varepsilon \,/\left(\pi^3 kT_e\right)^3}\,\exp\left(-\varepsilon \,/\, kT_e\right), \tag{2.1}$$

where k is a Boltzmann constant (if temperature is given in energy units, then $k = 1$ and can be omitted, we will follow this rule throughout the book). For this case, the **mean electron energy**, which is the first moment of the distribution function, is proportional to temperature in a traditional way:

$$\left\langle \varepsilon \right\rangle = \int_0^\infty \varepsilon f\left(\varepsilon\right) dE = \frac{3}{2} T_e. \tag{2.2}$$

Numerically, in most of the plasmas under consideration, the mean electron energy is from 1 to 5 eV.

### 2.1.2 Positive Ions

Atoms or molecules lose their electrons in ionization processes and form positive ions (Figure 2.1a). In extremely hot thermonuclear plasmas, the ions are multicharged, but in quasi-cold technological plasmas of interest – their charge is usually equal to +1e ($1.6\ 10^{-19}$ C). Ions are heavy particles, so commonly they cannot receive high energy directly from an electric field because of intensive collisional energy exchange with other plasma components. The collisional nature of the energy transfer results in the ion energy distribution function usually being not far from the quasi-equilibrium Maxwellian one (2.1), with the ion temperature $T_i$ close to the neutral gas temperature $T_o$. However, in some low-pressure discharges and sheaths, ion energy can be quite high.

One of the most important parameters of plasma generation is an ionization energy I, which is the energy needed to form a positive ion. Ionization requires quite large energy and as a rule defines the upper limit of microscopic energy transfer in plasma. Detailed information on ionization energies is available, for example, in V.N. Kondratiev, 1974. For some gases important in applications – the ionization energies are given in Table 2.1. Noble gases like He and Ne have the highest ionization energies I. Alkali metal vapors like $Li_2$ have the lowest value of I, so even a very small addition of such metals can dramatically stimulate ionization.

**TABLE 2.1**
**Ionization Energies for Different Atoms, Radicals and Molecules**

| | | | |
|---|---|---|---|
| $e + N_2 = N_2^+ + e + e$ | I = 15.6 eV | $e + O_2 = O_2^+ + e + e$ | I = 12.2 eV |
| $e + CO_2 = CO_2^+ + e + e$ | I = 13.8 eV | $e + CO = CO^+ + e + e$ | I = 14.0 eV |
| $e + H_2 = H_2^+ + e + e$ | I = 15.4 eV | $e + OH = OH^+ + e + e$ | I = 13.2 eV |
| $e + H_2O = H_2O^+ + e + e$ | I = 12.6 eV | $e + F_2 = F_2^+ + e + e$ | I = 15.7 eV |
| $e + H_2S = H_2S^+ + e + e$ | I = 10.5 eV | $e + HS = HS^+ + e + e$ | I = 10.4 eV |
| $e + SF_6 = SF_6^+ + e + e$ | I = 16.2 eV | $e + SiH_4 = SiH_4^+ + e + e$ | I = 11.4 eV |
| $e + UF_6 = UF_6^+ + e + e$ | I = 14.1 eV | $e + Cs_2 = Cs_2^+ + e + e$ | I = 3.5 eV |
| $e + Li_2 = Li_2^+ + e + e$ | I = 4.9 eV | $e + K_2 = K_2^+ + e + e$ | I = 3.6 eV |
| $e + CH_4 = CH_4^+ + e + e$ | I = 12.7 eV | $e + C_2H_2 = C_2H_2^+ + e + e$ | I = 11.4 eV |
| $e + CF_4 = CF_4^+ + e + e$ | I = 15.6 eV | $e + CCl_4 = CCl_4^+ + e + e$ | I = 11.5 eV |
| $e + H = H^+ + e + e$ | I = 13.6 eV | $e + O = O^+ + e + e$ | I = 13.6 eV |
| $e + He = He^+ + e + e$ | I = 24.6 eV | $e + Ne = Ne^+ + e + e$ | I = 21.6 eV |
| $e + Ar = Ar^+ + e + e$ | I = 15.8 eV | $e + Kr = Kr^+ + e + e$ | I = 14.0 eV |
| $e + Xe = Xe^+ + e + e$ | I = 12.1 eV | $e + N = N^+ + e + e$ | I = 14.5 eV |

$e + C_2H_5OH = e + C_2H_5OH^+ + 2e$ I = 10.5 eV $e + CH_3COOH = e + CH_3COOH^+ + 2e$ I = 10.4 eV
$e + C_6H_5OH = e + C_6H_5OH^+ + 2e$ I = 8.5 eV Naphthaldehyde I = 7.7 eV
$e + (C_2H_5)_2Cr = e + (C_2H_5)_2Cr^+ + 2e$ I = 5.5 eV 4-Methyl-Phenylene-Diamine I = 6.2 eV

### 2.1.3 Negative Ions

Electron attachment to atoms or molecules results in the formation of negative ions (Figure 2.1b). Their charge is equal to −1e (1.6 $10^{-19}$ C), attachment of another electron and formation of multicharged negative ions is actually impossible in the gas phase because of electric repulsion. Negative ions are heavy particles, so usually their energy balance is not due to the electric field, but rather due to collisional processes. The energy distribution functions for negative ions (the same as for positive ones) are not far from Maxwellian Eq. (2.1), with temperatures also close to those of the neutral gas.

EA can be defined as the energy release during attachment processes or as the bond energy between the attaching electron and the atom or molecule. EA is usually much lower than ionization energy I, even for very electronegative gases. Numerical values of EA for many different atoms, radicals, and molecules can be found in V.N. Kondratiev, 1974. Information for some gases of interest is presented in Table 2.2. Halogens and their compounds have the highest values of EA. Oxygen, ozone, and some oxides also are strongly electronegative substances with high EA. For this reason, nonthermal plasma generated in air usually contains a large number of negative ions.

### 2.1.4 Elementary Processes of the Charged Particles

Ionization of atoms and molecules by electron impact, electron attachment to atoms or molecules, and ion-molecular reactions (see Figure 2.1a–c) are examples of elementary plasma-chemical processes, reactive collisions accompanied by the transformation of elementary plasma particles. These elementary reactive collisions, as well as others, for example, electron–ion and ion-ion recombination, excitation and dissociation of neutral species by electron impact, relaxation of excited species, electron detachment and destruction of negative ions, photochemical processes – altogether determine plasma behavior. Elementary processes can be divided into two classes – elastic and nonelastic processes. The elastic collisions are those in which the internal energies of colliding particles do not change, therefore the total kinetic energy is conserved as well. Hence, these processes result only in scattering. Alternately collisions are inelastic. All elementary processes listed in previous

**TABLE 2.2**
**Electron Affinity for Different Atoms, Radicals and Molecules**

| | | | |
|---|---|---|---|
| $H^- = H + e$ | EA = 0.75 eV | $O^- = O + e$ | EA = 1.5 eV |
| $S^- = S + e$ | EA = 2.1 eV | $F^- = F + e$ | EA = 3.4 eV |
| $Cl^- = Cl + e$ | EA = 3.6 eV | $O_2^- = O_2 + e$ | EA = 0.44 eV |
| $O_3^- = O_3 + e$ | EA = 2.0 eV | $HO_2^- = HO_2 + e$ | EA = 3.0 eV |
| $OH^- = OH + e$ | EA = 1.8 eV | $Cl_2^- = Cl_2 + e$ | EA = 2.4 eV |
| $F_2^- = F_2 + e$ | EA = 3.1 eV | $SF_5^- = SF_5 + e$ | EA = 3.2 eV |
| $SF_6^- = SF_6 + e$ | EA = 1.5 eV | $SO_2^- = SO_2 + e$ | EA = 1.2 eV |
| $NO^- = NO + e$ | EA = 0.024 eV | $NO_2^- = NO_2 + e$ | EA = 3.1 eV |
| $NO_3^- = NO_3 + e$ | EA = 3.9 eV | $N_2O^- = N_2O + e$ | EA = 0.7 eV |
| $HNO_3^- = HNO_3 + e$ | EA = 2.0 eV | $NF_2^- = NF_2 + e$ | EA = 3.0 eV |
| $CH_3^- = CH_3 + e$ | EA = 1.1 eV | $CH_2^- = CH_2 + e$ | EA = 1.5 eV |
| $CF_3^- = CF_3 + e$ | EA = 2.1 eV | $CF_2^- = CF_2 + e$ | EA = 2.7 eV |
| $CCl_4^- = CCl_4 + e$ | EA = 2.1 eV | $SiH_3^- = SiH_3 + e$ | EA = 2.7 eV |
| $UF_6^- = UF_6 + e$ | EA = 2.9 eV | $PtF_6^- = PtF_6 + e$ | EA = 6.8 eV |
| $Fe(CO)_4^- = Fe(CO)_4 + e$ | EA = 1.2 eV | $TiCl_4^- = TiCl_4 + e$ | EA = 1.6 eV |

paragraphs are inelastic ones. Inelastic collisions result in the transfer of energy from the kinetic energy of colliding partners into internal energy. For example, processes of excitation, dissociation, and ionization of molecules by electron impact are inelastic collisions, including the transfer of high kinetic energy of plasma electrons into the internal degrees of freedom of the molecules.

In some instances, the internal energy of excited atoms or molecules can be transferred back into kinetic energy (in particular, into kinetic energy of plasma electrons). Such elementary processes are usually referred as **super-elastic collisions**. According to kinetic theory, the elementary processes can be described in terms of six main collision parameters: (1) cross section, (2) probability, (3) mean free path, (4) interaction frequency reaction rate, and (5) reaction rate coefficient (also termed reaction rate constant).

### 2.1.5 Fundamental Parameters of Elementary Processes

The most fundamental characteristics of all elementary processes are their cross sections. The **cross section of an elementary process** between two particles can be interpreted as an imaginary circle with area $\sigma$, moving together with one of the collision partners (see Figure 2.2a). If the center of the other collision partner crosses the circle, an elementary reaction occurs. The cross sections of elementary processes depend strongly on the energy of colliding species. If two colliding particles can be considered as hard elastic spheres of radii $r_1$ and $r_2$, their collisional cross section is equal to $\pi (r_1 + r_2)^2$. Obviously, the interaction radius and cross section can exceed the corresponding geometrical sizes because of long-distance nature of forces acting between electric charges and dipoles. Alternately, if only a few out of many collisions result in a chemical reaction, the cross section is considered to be less than the geometrical one. Typical sizes of atoms and molecules are of the order of 1–3 Å; therefore cross section of simple elastic collisions between

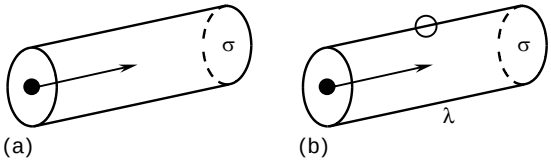



**FIGURE 2.2** (a) Cross section of an elementary process and (b) concept of the mean free path.

plasma electrons (energy 1–3 eV) and neutral particles is usually about $10^{-16}$–$10^{-15}$ cm$^2$. Cross sections of inelastic, endothermic, electron–neutral collisions are normally lower. When the electron energy is very large, the cross section can decrease because of a reduction of the interaction time. The ratio of the inelastic collision cross section to the corresponding cross section of an elastic collision under the same condition is often called the dimensionless **probability of the elementary process**.

**The mean free path** $\lambda$ of one collision partner A with respect to the elementary process A + B with another collision partner *B* can be calculated as

$$\lambda = 1/\left(n_B \sigma\right), \tag{2.3}$$

where $n_B$ is number density (concentration) of the particles B. During the mean free path $\lambda$, the particle A traverses the cylindrical volume $\lambda\sigma$. A reaction occurs if the cylindrical volume transversed by a particle A, contains at least one B-particle, which means $\lambda\sigma n_B = 1$ (see Figure 2.2b), the simple interpretation of Eq. (2.3) is the distance A travels before colliding. In the example of electrons colliding with neutrals, the electron mean free path with respect to elastic collisions with neutrals at atmospheric pressure and $n_B = 3*10^{19}$ cm$^{-3}$ is approximately $\lambda = 1\mu$.

**The interaction frequency** $\nu$ of one collision partner A (e.g., electrons moving with velocity v) with the other collision partner B (e.g., a heavy neutral particle) can be defined as $v/\lambda$ or taking into account Eq. (2.3) as

$$\nu_A = n_B <\sigma v> \tag{2.4}$$

This relation should be averaged taking into account the velocity distribution function f(v) and the dependence of the cross section $\sigma$ on the collision partners' velocity. When the collision partners' velocity can be attributed mostly to one light particle (e.g., electron) Eq. (2.4) can be rewritten as:

$$\nu_A = n_B \int f\left(v\right)\sigma\left(v\right)vdv = <\sigma v> n_B. \tag{2.5}$$

Numerically, for the abovementioned example of atmospheric pressure, electron neutral, elastic collisions, the interaction frequency is approximately $\nu = 10^{12}$ 1/s.

### 2.1.6 The Reaction Rate Coefficients

The number of elementary processes, which occur in unit volume per unit time is called the elementary reaction rate. This concept can be used for any kind of reaction: mono-molecular, bimolecular, and three-body reactions. For bimolecular processes A + B, the reaction rate can be calculated by multiplying the interaction frequency of partner A with partner B - "$\nu_A$" and number of particles A in the unit volume (which is their number density- $n_A$). Thus, $w = \nu_A n_A$. Taking into account Eq. (2.5), this can be rewritten as:

$$w_{A+B} = \left\langle \sigma v \right\rangle n_A n_B. \tag{2.6}$$

The factor $<\sigma v>$ is termed the **reaction rate coefficient** (or reaction rate constant), one of the most useful concepts of plasma chemical kinetics. For bimolecular reactions, it can be calculated as

$$k_{A+B} = \int f\left(v\right)\sigma\left(v\right)vdv = <\sigma v> \tag{2.7}$$

In contrast to the reaction cross section σ, which is a function of the partners' energy, the reaction rate coefficient k is an integral factor, which includes information on energy distribution functions and depends on temperatures or mean energies of the collision partners. Equation (2.7) establishes the relation between micro-kinetics (which is concerned with elementary processes) and macro-kinetics (which takes into account real energy distribution functions). Numerically, for the above-considered example of electron neutral, elastic binary collisions, the reaction rate coefficient is approximately $k = 3 * 10^{-8}$ cm$^3$/s.

The concept of reaction rate coefficients can be applied not only for bimolecular processes Eq. (2.7) but also for mono-molecular (A → products) and three-body processes (A + B + C→ products). Then, the reaction rate is proportional to a product of concentrations of participating particles, and the reaction rate coefficient is just a coefficient of the proportionality:

$$w_A = k_A n_A \ (a); \quad w_{A+B+C} = k_{A+B+C} n_A n_B n_C \ (b). \tag{2.8}$$

Obviously, in these cases Eq. (2.8), the reaction rate coefficients cannot be calculated using Eq. (2.7), they even have different dimensions. Also, for mono-molecular processes Eq. (2.8a), the reaction rate coefficient can be interpreted as the reaction frequency.

### 2.1.7 Elementary Elastic Collisions of Charged Particles

Elastic collisions (or, elastic scattering) do not result in a change of chemical composition or excitation level of the colliding partners and for this reason, they are not able to influence directly plasma-chemical processes. On the other hand, due to their high values of cross sections, elastic collisions are responsible for kinetic energy and momentum transfer between colliding partners. This leads to their important role in physical kinetics – in plasma conductivity, drift and diffusion, absorption of electromagnetic energy, and in the evolution of energy distribution functions. The typical value for cross sections of the elastic, electron–neutral collisions for electron energy of 1–3 eV is approximately $\sigma = 10^{-16}$–$10^{-15}$ cm$^2$; the reaction rate coefficient approximately $k = 3*10^{-8}$ cm$^3$/s. For the elastic, ion-neutral collisions at room temperature, the typical value of cross section is about $\sigma = 10^{-14}$ cm$^2$; and reaction rate coefficient of about $k = 10^{-9}$ cm$^3$/s.

Electron–electron, electron–ion, and ion-ion scattering processes are the so-called **Coulomb collisions**. While their cross sections are quite large with respect to those of collisions with neutral partners, these are relatively infrequent processes in discharges with a low degree of ionization. An important feature of the Coulomb collisions is a strong dependence of their cross sections on the kinetic energy of colliding particles. This effect can be demonstrated by a simple analysis, illustrated in Figure 2.3. Here, two particles have the same charge and for simplicity, one collision partner is considered at rest. Scattering takes place if the Coulomb interaction energy (order of $U \sim q^2/b$, where b is the impact parameter) is approximately equal to the kinetic energy ε of a collision partner. Then, the impact parameter $b \sim q^2/\varepsilon$, and the reaction cross section σ can be estimated as $\pi b^2$ and

$$\sigma(\varepsilon) \sim \pi q^4/\varepsilon^2 \tag{2.9}$$

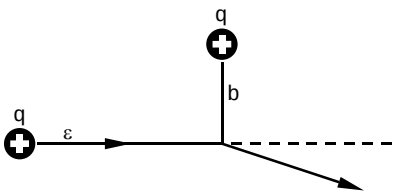



**FIGURE 2.3** Coulomb collisions illustration.

The electron–electron scattering cross sections at room temperature is about 1000 times greater than those at electron temperature of 1 eV. Similar consideration of the charged particle scattering on neutral molecules with a permanent dipole momentum (interaction energy $U \sim 1/r^2$) and an induced dipole momentum (interaction energy $U \sim 1/r^4$) gives, respectively, $\sigma(\varepsilon) \sim 1/\varepsilon$ and $\sigma(\varepsilon) \sim 1/\varepsilon^{1/2}$.

Energy transfer during elastic collisions is possible only as a transfer of kinetic energy. The average fraction $\gamma$ of kinetic energy transferred from one particle of mass m to another one of mass M is:

$$\gamma = 2mM / \left(m + M\right)^2 \tag{2.10}$$

In an elastic collision of electrons with heavy neutrals or ions $m << M$, and hence $\gamma = 2\ m/M$, which means the fraction of transferred energy is negligible ($\gamma \sim 10^{-4}$). One can find detailed fundamental consideration of the elementary elastic collisions of charged species as well as corresponding experimental data in the books of B.M. Smirnov (1981), E.W. McDaniel (1964, 1989), and Massey et al (1978)

## 2.2 IONIZATION PROCESSES

The key process in plasma is ionization because it is responsible for plasma generation – the birth of new electrons and positive ions. The simplest ionization process – the ionization by electron impact – is illustrated in Figure 2.1a. In general, all ionization processes can be subdivided into five groups:

The first group, **direct ionization by electron impact**, includes the ionization of neural neutrals and preliminary not-excited atoms, radicals, or molecules by an electron, whose energy is sufficiently high enough to provide the ionization act in one collision. These processes are the most important in cold or nonthermal discharges, where electric fields and hence electron energies are quite high, but the level of excitation of neutral species is relatively moderate. The second group, **stepwise ionization by electron impact**, includes the ionization of preliminary excited neutral species. These processes are important mainly in thermal or energy-intense discharges when the ionization degree (ratio of number densities of electrons and neutrals) and the concentration of highly excited neutral species is quite high. The third group is the **ionization by a collision of heavy particles**. Such processes can take place during ion-molecular or ion-atomic collisions, as well as in collisions of electronically or vibrationally excited species when the total energy of the collision partners exceeds the ionization potential. The chemical energy of the colliding neutral species can be also contributed to ionization via the so-called associative ionization. The fourth group is the **photoionization,** where neutral collisions with photons result in the formation of an electron–ion pair. Photoionization is mainly important in thermal plasmas and some mechanisms of propagation of nonthermal discharges.

### 2.2.1 The Direct Ionization by Electron Impact

This process takes place as a result of the interaction of an incident electron, having a high energy $\varepsilon$, with a valence electron of a preliminary neutral atom or molecule. The act of ionization occurs when energy $\Delta\varepsilon$ transferred between them exceeds the ionization potential I (see Section 2.1.2). Physical picture and good quantitative formulas can be derived from the following classical model, first introduced by Thomson, 1912. The Thomson model supposes that the valence electron is at rest and also neglects the interaction of the two colliding electrons with the rest part of the initially neutral particle. The differential cross section of the incident electron scattering with energy transfer $\Delta\varepsilon$ to the valence electron can be defined by the Rutherford formula:

$$d\sigma_i = \frac{1}{\left(4\pi\varepsilon_0\right)^2} \frac{\pi e^4}{\varepsilon \left(\Delta\varepsilon\right)^2} d\left(\Delta\varepsilon\right) \tag{2.11}$$

Integration of the differential expression Eq. (2.11) over $\Delta\varepsilon$, taking into account that for ionization acts, the transferred energy should exceed the ionization potential $\Delta\varepsilon \geq I$, gives

$$\sigma_i = \frac{1}{\left(4\pi\varepsilon_0\right)^2}\frac{\pi e^4}{\varepsilon}\left(1/I - 1/\varepsilon\right). \tag{2.12}$$

This relation describes the direct ionization cross section $\sigma_i$ as a function of an incident electron energy $\varepsilon$ and it is known as the **Thomson formula**. Obviously, Eq. (2.12) should be multiplied, in general, by number of valence electrons $Z_v$. According to the Thomson formula, the direct ionization cross section is growing linearly near the threshold of the elementary process $\varepsilon = I$. In the case of high electron energies $\varepsilon >> I$, the Thomson cross section (2.12) decreases as $\sigma_i \sim 1/\varepsilon$. Quantum mechanics gives a more accurate asymptotic approximation for the high energy electrons $\sigma_i \sim \ln\varepsilon/\varepsilon$. When $\varepsilon = 2I$, the Thomson cross section reaches the maximum value:

$$\sigma_i^{\max} = \frac{1}{\left(4\pi\varepsilon_0\right)^2}\frac{\pi e^4}{4I^2} \tag{2.13}$$

The Thomson formula can be rewritten more precisely, taking into account kinetic energy $\varepsilon_v$ of the valence electron (B.M. Smirnov, 1981):

$$\sigma_i = \frac{1}{\left(4\pi\varepsilon_0\right)^2}\frac{\pi e^4}{\varepsilon}\left(1/\varepsilon - 1/I + \frac{2\varepsilon_v}{3}\left(\frac{1}{I^2} - \frac{1}{\varepsilon^2}\right)\right) \tag{2.14}$$

The Thomson formula (2.12) can be derived from Eq. (2.14) by assuming that the valence electron is at rest and $\varepsilon_v = 0$. A useful variation of the Thomson formula can be obtained, assuming the valence electron interaction with the rest of the atom as a Coulomb interaction. Then, according to classical mechanics $\varepsilon_v = I$ and Eq. (2.14) gives another relation for the direct ionization cross section, which takes into account motion of valence electrons:

$$\sigma_i = \frac{1}{\left(4\pi\varepsilon_0\right)^2}\frac{\pi e^4}{\varepsilon}\left(5/3I - 1/\varepsilon - 2I/3\varepsilon^2\right) \tag{2.15}$$

Equations (2.12) and (2.15) describe the direct ionization in a similar way, but the latter predicts a twice larger cross section $\sigma_i(\varepsilon)$ maximum ($\sigma_i^{\max} \sim \pi e^4/2I^2(4\pi\varepsilon_0)^2$) at a slightly lower electron energy $\varepsilon_{\max} = 1.85\ I$. All the discussed relations for the direct ionization cross section can be generalized as:

$$\sigma_i = \frac{1}{\left(4\pi\varepsilon_0\right)^2}\frac{\pi e^2}{I^2}Z_v f\left(\frac{\varepsilon}{I}\right), \tag{2.16}$$

where $Z_v$ is a number of valence electrons, and $f(\varepsilon/I) = f(x)$ is a general function, which is common for all atoms. Thus, for the Thomson formula (2.12):

$$f\left(x\right) = \frac{1}{x} - \frac{1}{x^2} \tag{2.17}$$

Equation (2.16) is in a good agreement with experimental data for different atoms and molecules (B.M. Smirnov, 1981) if:

$$\frac{10\left(x-1\right)}{\pi\left(x+0.5\right)\left(x+8\right)} < f\left(x\right) < \frac{10\left(x-1\right)}{\pi x\left(x+8\right)} \tag{2.18}$$

Formulas for practical estimations are discussed in Barnett (1989). The semi-empirical formula (2.18) is quite useful for numerical calculations, if the relation $\sigma_i(\varepsilon)$ is unknown experimentally.

### 2.2.2 The Direct Ionization Rate Coefficient

The ionization rate coefficient $k_I$ (2.7) can be calculated by the integration of the cross section $\sigma_i(\varepsilon)$ over the electron energy distribution function. Electron energy distribution function is determined by electron temperature $T_e$ (or mean energy (2.2)), therefore the ionization rate coefficient $k_I(T_e)$ is also a function of $T_e$. When the ionization rate coefficient is known, the rate of direct ionization by electron impact $w_{ion}$ can be found as:

$$w_{ion} = k_I\left(T_e\right) n_e\, n_0. \tag{2.19}$$

In this formula, $n_e$ is the concentration (number density) of electrons, and $n_0$ is the concentration of neutral gas atoms or molecules. The ionization potential I is usually much greater than the mean electron energy. For this reason, the ionization rate coefficient is very sensitive to the electron energy distribution function. If $T_e << I$, only a small group of electrons can have energy exceeding the ionization potential. Then integrating $\sigma_i(\varepsilon)$ into Eq. (2.7), the linear part of $\sigma_i(\varepsilon)$ near the threshold of ionization $\varepsilon = I$ is sufficient:

$$\sigma_i^{threshold}\left(\varepsilon\right) = \frac{1}{\left(4\pi\varepsilon_0\right)^2} Z_v \frac{\pi e^4}{I^2}\left(\varepsilon/I - 1\right) = \sigma_0\left(\varepsilon/I - 1\right) \tag{2.20}$$

Here, $\sigma_0 = Z_v \pi e^4 / I^2 (4\pi\varepsilon_0)^2$ is the geometrical atomic cross section. After integrating Eq. (2.7), the direct ionization rate coefficient can be presented as:

$$k_i\left(T_e\right) = \sqrt{8T_e / \pi m}\,\sigma_0 \exp\left(-I/T_e\right) \tag{2.21}$$

For numerical estimations, one can take the cross section $\sigma_0$ for molecular nitrogen, $10^{-16}$ cm$^2$, and for argon $3*10^{-16}$ cm$^2$. Electron temperature is presented here in energy units, so the Boltzmann coefficient k = 1. Some numerical data on the electron impact direct ionization for different molecular gases – $CO_2$, $H_2$, $N_2$ e.a. – are presented in Figure 2.4 as a function of reduced electric field $E/n_0$, which is the ratio of the electric field over neutral gas concentration.

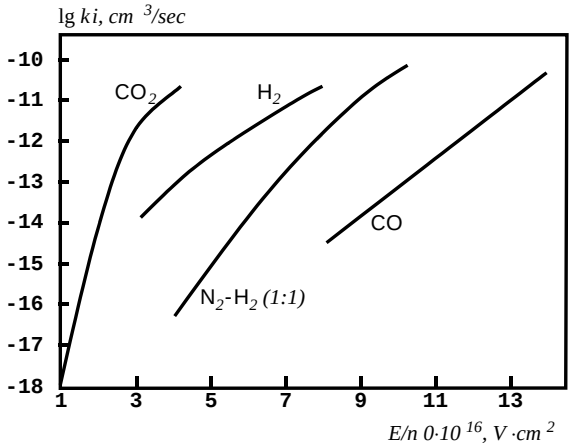



**FIGURE 2.4** Ionization rate coefficient in molecular gases as a function of reduced electric field.

### 2.2.3 Peculiarities of Dissociation of Molecules by Electron Impact. The Frank-Condon Principle and the Process of Dissociative Ionization

First, let us consider a process of non-dissociative ionization of molecules by direct electron impact, taking as an example – the case of ionization of diatomic molecules AB:

$$e + AB \rightarrow AB^+ + e + e \tag{2.22}$$

This process takes place predominantly when the electron energy exceeds the ionization potential not very essentially. It can be described in the first approximation by the Thomson approach (Section 2.2.1). One can see some peculiarities of ionization of molecules by electron impact, using the illustrative potential energy curves for AB and $AB^+$, shown in Figure 2.5. The fastest internal motion of atoms inside of molecules is molecular vibration. However, even molecular vibrations have typical times of $10^{-14}$–$10^{-13}$ s, which is much longer, than the interaction time between plasma electrons and the molecules $a_0/v_e \sim 10^{-16}$–$10^{-15}$ s (here $a_0$ is the atomic unit of length, $v_e$ is the mean electron velocity). This means, that all kinds of electronic excitation processes under consideration, induced by electron impact (including the molecular ionization, see Figure 2.5), are much faster, than all kinds of atomic motions inside of the molecules. As a result, all of the atoms inside molecules can be considered as frozen during the process of electronic transition, stimulated by electron impact. This fact is known as the **Frank-Condon Principle.**

According to the Frank-Condon Principle, the processes of collisional excitation and ionization of molecules are presented in Figure 2.5 by vertical lines (inter-nuclear distances are constant). This means that the non-dissociative ionization process Eq. (2.22) usually results in the formation of a vibrationally excited ion $(AB^+)^*$ and requires a little more energy, than corresponding atomic ionization. The phenomena can be described, using the **Frank-Condon factors**. When the electron energy is relatively high and essentially exceeds the ionization potential, the dissociative ionization process can take place:

$$e + AB \rightarrow A + B^+ + e + e \tag{2.23}$$

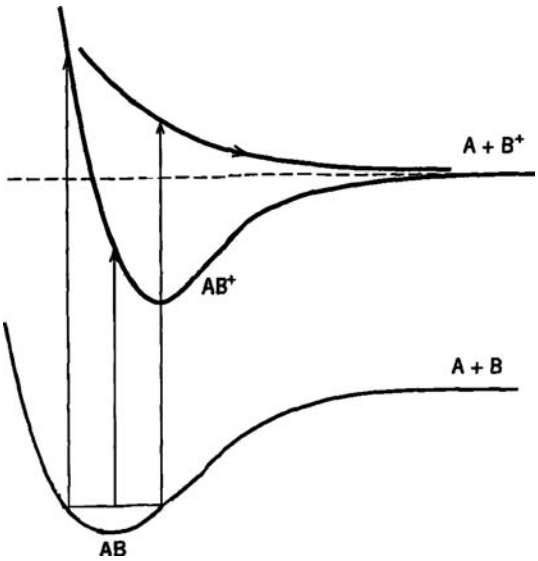



**FIGURE 2.5** Molecular and ionic terms illustrating dissociative ionization.

This ionization process corresponds to electronic excitation into a repulsive state of ion $(AB^+)^*$, followed by the decay of this molecular ion. This is also illustrated by the vertical line in Figure 2.5. The energy threshold for the dissociative ionization is essentially greater than for the non-dissociative one.

### 2.2.4 Stepwise Ionization by Electron Impact

If the number density of electrons and hence the concentration of excited neutral species are sufficiently high, the energy I, necessary for ionization, can be provided in two different ways: (1) by the energy of plasma electrons and (2) by the preliminary electronic excitation of neutrals, which is called the stepwise ionization. The total energy needed for ionization in these two processes is sure the same, but which process is more preferable and which contributes the most in the total rate of ionization? The answer is unambiguous: if the level of electronic excitation is sufficiently high enough, the stepwise ionization is much faster, than the direct one, because the statistical weight of electronically excited neutrals is greater than that of free plasma electrons. In other words, when $T_e << I$, the probability of obtaining the ionization energy I is much lower for free plasma electrons (direct ionization) than for excited atoms and molecules. In contrast with the direct ionization, the stepwise process includes several steps, several electron impacts to provide the ionization act. The first electron – neutral collisions result in the preparation of the highly excited species, and then a final collision with relatively low energy electron provides the actual ionization act.

In the thermodynamic equilibrium, the ionization processes $e + A \rightarrow A^+ + e + e$ are reversed with respect to three-body recombination $A^+ + e + e \rightarrow A^* + e \rightarrow A + e$, which is proceeding through a set of excited states. According to the principle of detailed equilibrium, it means, that the ionization $e + A \rightarrow A^+ + e + e$ should go through the set of electronically excited states as well, which means the ionization should be a stepwise process. The stepwise ionization rate coefficient $k_i^s$ can be found by summation of the partial rate coefficients $k_i^{s,n}$, corresponding to the n-th electronically excited state, over all states of excitation, taking into account their concentrations:

$$k_i^s = \sum_n k_i^{s,n} N_n(\varepsilon_n) / N_0 \tag{2.24}$$

To calculate the maximum stepwise ionization rate (B.M. Smirnov, 1981), let us suppose, that the electronically excited atoms and molecules are in quazi quasi-equilibrium with plasma electrons, and that, the excited species have an energy distribution function, corresponding to the Boltzmann law with the electron temperature $T_e$:

$$N_n = \left(g_n / g_0\right) N_0 \exp\left(-\varepsilon_n / T_e\right). \tag{2.25}$$

Here $N_n$, $g_n$, and $\varepsilon_n$ are number density, statistical weight, and energy (with respect to ground state) of the electronically excited atoms, radicals, or molecules; the index "n" is the principal quantum number and actually it shows the particle excitation level. From statistical thermodynamics, the statistical weight of an excited particle $g_n = 2g_i n^2$, where $g_i$ is the statistical weight of ion. $N_0$ and $g_0$ – are the concentration and statistical weight of ground state particles. Once more it is noted that the Boltzmann constant can be taken as $k = 1$ and for this reason omitted if the temperature is given in energy units. We'll follow this rule, in most cases later on in this book.

Typical energy transfer from a plasma electron to an electron, sitting on an excited atomic level is about $T_e$. This means that excited particles with energy $\varepsilon_n = I - T_e$ make the most important contribution to the sum Eq. (2.24). Taking into account that $I_n \sim 1/n^2$, the number of states

with energy about $\varepsilon_n = I - T_e$ and ionization potential about $I_n = T_e$ has an order of n. Thus, from Eqs. (2.24) and Eqs. (2.25):

$$k_i^s \approx \frac{g_i}{g_0} n^3 \langle \sigma v \rangle \exp\left(-I/T_e\right) \tag{2.26}$$

The cross section σ in Eq. (2.26), corresponding to energy transfer (about $T_e$) between electrons, can be estimated as $e^4/T_e^2(4\pi\varepsilon_0)^2$, velocity $v \sim \sqrt{T_e/m}$ and the quantum number can be taken from:

$$I_n \approx \frac{1}{\left(4\pi\varepsilon_0\right)^2} me^4 / \hbar n^2 \approx T_e \tag{2.27}$$

As a result, the stepwise ionization rate can be presented based on Eqs. (2.26) and (2.27) as:

$$k_i^s \approx \frac{g_i}{g_0} \frac{1}{\left(4\pi\varepsilon_0\right)^5} \left(me^{10} / T_e^3\right) \exp\left(-I/T_e\right) \hbar^3 \tag{2.28}$$

Here $\hbar$ is the Planck's constant. In general, Eq. (2.28) is quite convenient for numerical estimations of the stepwise ionization rate in thermal plasmas, when the electronically excited species are in quasi-equilibrium with the gas of plasma electrons. Also, the expression Eq. (2.28) for stepwise ionization is in a good agreement (in the framework of the principle of detailed equilibrium) with the rate of the reverse reaction – the three-body recombination $A^+ + e + e \leftrightarrow A^* + e \rightarrow A + e$.

Comparing direct ionization Eq. (2.21) with the stepwise ionization Eq. (2.28), one can see, that the second one can be much faster, because of the high large statistical weight of excited species, involved in the stepwise ionization. The ratio of rate coefficients for these two competing mechanisms of ionization can be derived from Eqs. (2.21) and (2.18):

$$\frac{k_i^s\left(T_e\right)}{k_i\left(T_e\right)} \approx \frac{g_i a_0^2}{g_0 \sigma_0} \left( \frac{1}{\left(4\pi\varepsilon_0\right)^2} me^4 / \hbar T_e \right)^{7/2} \approx \left( \frac{I}{T_e} \right)^{7/2} \tag{2.29}$$

Here $a_0$ is the atomic unit of length. We have taken into account in this relation, - the estimations for geometric collisional cross section $\sigma_0 \sim a_0^2$, and for ionization potential: $I \approx \frac{1}{\left(4\pi\varepsilon_0\right)^2} me^4 / \hbar^2$. For typical discharges with $I/T_e \sim 10$, the stepwise ionization can be $10^3$–$10^4$ times faster, than the direct one. This takes place only in the case of high electron concentration $n_e$, and high electronic excitation frequency $k_{en}n_e$, which results in the quasi-equilibrium Eq. (2.25) between electronically excited species and plasma electrons. Usually, it can be applied only for quasi-equilibrium plasmas of thermal discharges.

To estimate the contribution of stepwise ionization in non-equilibrium discharges let us make some corrections in Eq. (2.29). If deactivation of the excited species because of radiation, collisional relaxation and losses on the walls has a characteristic time $\tau_n^*$ shorter than excitation time $1/k_{en}n_e$, the concentration of excited species will be lower, than the equilibrium one Eq. (2.25). Equation (2.25) can be corrected by taking into account electron concentration and the non-equilibrium losses of the excited species:

$$N_n \approx \frac{g_n}{g_0} N_0 \frac{k_{en} n_e \tau_n^*}{k_{en} n_e \tau_n^* + 1} \exp\left(-\varepsilon_n / T_e\right). \tag{2.30}$$

Then the stepwise ionization rate in non-equilibrium discharges can be rewritten as a function of electron temperature as well as electron concentration:

$$k_i^s\left(T_e,n_e\right) \approx k_i\left(T_e\right)\frac{k_{en}n_e\tau_n^*}{k_{en}n_e\tau_n^*+1}\left(\frac{I}{T_e}\right)^{\frac{7}{2}}. \tag{2.31}$$

One can see, that when the electron concentration is sufficiently high, this expression corresponds to Eq. (2.29) for the stepwise ionization rate in quasi-equilibrium plasmas.

### 2.2.5 Ionization by High Energy Electron Beams

For many important modern applications, plasma is generated not by electric discharges, but by means of the high energy electron beams. The ionization in this case can be more homogeneous and provide a generation of uniform nonthermal plasmas even in atmospheric pressure systems. The electron beams can be also effectively combined with the electric field in non-self-sustained discharges, where electron beams providing ionization, and the energy consumption is mostly due to the electric field. Peculiarities of ionization, in this case, are due to high energy of electrons in the beams, which usually varies from 50 keV to 1–2 MeV. Typical energy losses of the beams in atmospheric pressure air are of about 1 MeV/m per 1m. The beams with electron energies more than 500 keV are referred to as relativistic electron beams because these energies exceed the relativistic electron energy at rest ($E = mc^2$). In the frameworks of the Born approximation (Landau, 1997), electron energy losses per unit length dE/dx can be evaluated by the non-relativistic Bethe-Bloch formula:

$$-\frac{dE}{dx} = \frac{2\pi Ze^4}{\left(4\pi\varepsilon_0\right)^2 mv^2}n_0 \ln\frac{2mEv^2}{I^2}. \tag{2.32}$$

Here Z is an atomic number of neutral particles, providing the beam stopping; $n_0$ is their number density; v is the stopping electron velocity. In the case of relativistic electron beams with the electron energies 0.5–1 MeV, the energy losses going to neutral gas ionization can be numerically calculated by the relation:

$$-\frac{dE}{dx} = 2*10^{-22}n_0 Z \ln\frac{183}{Z^{1/3}}, \tag{2.33}$$

where energy losses dE/dx are expressed in MeV/cm, and concentration of neutral particles $n_0$ is expressed– in $cm^{-3}$. Equation (2.33) can be rewritten in terms of effective ionization rate coefficient $k_i^{eff}$ for relativistic electrons:

$$k_i^{eff} \approx 3*10^{-10}\left(\mathrm{cm}^3/\mathrm{sec}\right)Z\ln\frac{183}{Z^{1/3}} \tag{2.34}$$

Numerically, this ionization rate coefficient is about $10^{-8}$–$10^{-7}$ $cm^3/s$. The rate of ionization by relativistic electron beams can be expressed in this case as a function of the electron beam concentration $n_b$ or the electron beam current density $j_b$:

$$q_e = k_i^{eff}n_b n_0 \approx k_i^{eff}\frac{1}{ec}n_0 j_b, \tag{2.35}$$

where c is the sped of light.

### 2.2.6 Photoionization Processes

Though ionization processes are mostly induced by electron impact, ionization also can be also provided by interaction with high energy photons. The process of photoionization of a neutral particle A with ionization potential I (in eV) by a photon ħω with wavelength λ can be illustrated as:

$$\hbar\omega + A \rightarrow A^{+} + e, \lambda < \frac{12,400}{I\left(eV\right)}\text{A} \tag{2.36}$$

To provide ionization, the UV-photon wavelength should be quite low, usually below 100 nm. However, to provide effective ionization of preliminary excited atoms and molecules, photon energy can be lower. The photoionization cross section increases sharply from zero at the threshold energy Eq. (2.36) to quite high values, up to the geometrical cross section. Specific numerical values of the photoionization cross sections (near the threshold of the process) are presented in Table 2.3.

Though the cross sections shown in Table 2.3 are quite high, the contribution of the photoionization process is usually not very significant, because of the low concentration of high energy photons in most situations. However, in the following examples the photoionization plays a very essential role mostly by rapidly supplying seed electrons for the following ionization by electron impact:

1. streamer propagation in nonthermal discharges, where photoionization supplies seed electrons to start electron avalanches;
2. propagation of both nonthermal and thermal discharges in fast flow including supersonic one, where other mechanisms of discharge propagation are too slow;
3. preliminary gas ionization by ultraviolet radiation in non-self-sustained discharges, where UV-radiation is a kind of replacement of the relativistic electron beams.

### 2.2.7 The Ionization by Collisions of Heavy Particles, Adiabatic Principle, and Parameter Massey

An electron with kinetic energy slightly exceeding the ionization potential is quite effective to perform the ionization act. However, that is not true for ionization by collisions of heavy particles-ions and neutrals. Even when heavy particles have enough kinetic energy, the ionization probability is very low because their velocities are much less than those of electrons in atoms. This effect is a reflection of a

**TABLE 2.3**
**The Photoionization Cross Sections**

| Atoms or Molecules | Wavelength λ, in A | Cross sections, in cm$^2$ |
|---|---|---|
| Ar | 787 | $3.5*10^{-17}$ |
| $N_2$ | 798 | $2.6*10^{-17}$ |
| N | 482 | $0.9*10^{-17}$ |
| He | 504 | $0.7*10^{-17}$ |
| $H_2$ | 805 | $0.7*10^{-17}$ |
| H | 912 | $0.6*10^{-17}$ |
| Ne | 575 | $0.4*10^{-17}$ |
| O | 910 | $0.3*10^{-17}$ |
| $O_2$ | 1020 | $0.1*10^{-17}$ |
| Cs | 3185 | $2.2*10^{-19}$ |
| Na | 2412 | $1.2*10^{-19}$ |
| K | 2860 | $1.2*10^{-20}$ |

general principle of particle interaction. A slow motion is "adiabatic", reluctant to transfer energy to a fast motion. The **adiabatic principle** can be explained in terms of relations between low interaction frequency $\omega_{int} = \alpha v$ (reverse time of interaction between particles) and high frequency of electron transfer in atom $\omega_{tr} = \Delta E\hbar$ Here $1/\alpha$-is a characteristic size of the interacting neutral particles, v is their velocity, $\Delta E$ is a change of electron energy in atom during the interaction, $\hbar$ is the Planck's constant. Only fast Fourier components of the slow interaction potential between particles with frequencies about $\omega_{tr} = \Delta E/\hbar$, provide the energy transfer between the interacting particles. The relative contribution of these fast Fourier components is very low if $\omega_{tr} >> \omega_{int}$, numerically it is approximately: $\exp(-\omega_{tr}/\omega_{int})$. As a result, the probability $P_{EnTr}$ and cross sections of energy transfer processes (including the ionization process under consideration) are usually proportional to the so-called Massey parameter:

$$P_{EnTr} \propto \exp\left(-\frac{\omega_{tr}}{\omega_{\text{int}}}\right) \propto \exp\left(-\frac{\Delta E}{\hbar \alpha v}\right) = \exp\left(-P_{Ma}\right) \tag{2.37}$$

Here $P_{Ma} = \Delta E\hbar\alpha v$ is the **adiabatic Massey parameter**. If $P_{Ma} >> 1$ the process of energy transfer is adiabatic, and its probability is exponentially low, which occurs in the collisions of energetic heavy neutrals and ions. To get the Massey parameter close to one and eliminate the adiabatic prohibition for ionization, the kinetic energy of the colliding heavy particle has to be about 10–100 KeV, which is about three orders of magnitude greater than ionization potential. Though the kinetic energy of heavy particles in the ground state is ineffective for ionization, the situation can be different if they are electronically excited. If the total electron excitation energy of the colliding heavy particles is close to the ionization potential of one of them, the resonant energy transfer and effective ionization act can take place. Such non-adiabatic ionization processes occurring in the collision of heavy particles will be illustrated below by two specific examples: the Penning ionization effect and associative ionization.

### 2.2.8 The Penning Ionization Effect and Process of Associative Ionization

If electron excitation energy of a metastable atom $A^*$ exceeds the ionization potential of another atom B, their collision can lead to an act of ionization, the so-called **Penning ionization**. The Penning ionization usually takes place through the intermediate formation of an excited molecule (in the state of auto-ionization), and cross sections of the process can be very large. The cross sections for the Penning ionization of $N_2$, $CO_2$, Xe, and Ar by metastable helium atoms $He(2^3S)$ with excitation energy 19.8 eV reach gas-kinetic values of $10^{-15}$ cm$^2$. Similar cross sections can be attained in collisions of metastable neon atoms (excitation energy 16.6 eV) with argon atoms (ionization potential 15.8 eV). Exceptionally high cross section $1.4*10^{-14}$ cm$^2$ (B.M. Smirnov, 1974) can be attained in the Penning ionization of mercury atoms (ionization potential 10.4 eV) by collisions with the metastable helium atoms $He(2^3S$, 19.8 eV). If the total electron excitation energy of colliding particles is not sufficient, an ionization process is possible nevertheless when heavy species stick to each other, forming a molecular ion, and hence their bonding energy can be also contributed to the ionization act. Such a process is called **the associative ionization.** The process differs from the Penning ionization only by the stability of the molecular ion product. A good example of the associative ionization is the process, involving the collision of two metastable mercury atoms:

$$Hg\left(6^3P_1, E = 4.9\,eV\right) + Hg\left(6^3P_0, E = 4.7\,eV\right) \rightarrow Hg_2^+ + e \tag{2.38}$$

The total electron excitation energy here, 9.6 eV, is less than the ionization potential of mercury atom (10.4 eV), but higher than that for $Hg_2$ molecule. This is the main "trick" of the associative ionization. Cross sections of the associative ionization (similarly to the Penning ionization) can be high and close to the gas-kinetic one ($10^{-15}$ cm$^2$). The associative ionization is a reverse process with

respect to dissociative recombination $e + AB^+ \rightarrow A^* + B$, the main recombination mechanism for molecular ions. The associative ionization is effective only for such excited species, which can be produced during the dissociative recombination. The relation between the cross sections of associative ionization $\sigma_{ai}$ and dissociative recombination $\sigma_r^{ei}$ can be derived (B.M. Smirnov, 1981) from the principle of detailed equilibrium as:

$$\sigma_{ai}\left(v_{rel}\right) = \sigma_r^{ei}\left(v_e\right)\frac{m^2 v_e^2}{\mu^2 v_{rel}^2}\frac{g_e g_{AB^+}}{g_{A^*} g_B} \tag{2.39}$$

In this relation: $v_{rel}$ and $v_e$, are respectively, the relative velocity of heavy particles and the electron velocity, $\mu$ is the reduced mass of heavy particles, g are statistical weights. Ionization (and in particular the associative ionization) can also occur in the collision of vibrationally excited molecules. Cross sections and reaction rate coefficients for such processes are very low because of low Frank-Condon factors in this case. Nevertheless, these processes can make a notable contribution to ionization in the absence of an electric field (V. Rusanov, A. Fridman, 1984), like in the following process:

$$N_2^*\left(\,^1\Sigma_g^+, v_1 \approx 32\right) + N_2^*\left(\,^1\Sigma_g^+, v_2 \approx 32\right) \rightarrow N_4^+ + e \tag{2.40}$$

While this associative ionization has enough energy accumulated in the nitrogen molecules with 32 vibrational quanta each, but anyway the reaction rate of the process is relatively low: $10^{-15}$ exp (–2000 K/ T), $cm^3/s$ (I.A. Adamovich et al., 1993, E. Plouges et al., 2001).

## 2.3 MECHANISMS OF ELECTRON LOSSES: THE ELECTRON–ION RECOMBINATION

The variety of channels of charged particle losses can be subdivided into three major groups. The first group includes different types of **electron–ion recombination processes**, in which collisions of the charged particles in a discharge volume lead to their mutual neutralization. These exothermic processes require consuming in some manner the large release of recombination energy. Dissociation of molecules, radiation of excited particles, or three-body collisions can provide the consumption of the recombination energy. All these different mechanisms of the electron–ion recombination will be discussed in this section.

Electron losses because of their sticking to neutrals and formation of negative ions (see Figure 2.1b) form the second group of volumetric losses, **electron attachment processes**. These processes are often responsible for the balance of charged particles in such electro-negative gases as oxygen (and for this reason air), $CO_2$ (because of formation of $O^-$), different halogens, and their compounds. Reverse processes of an electron release from a negative ion are called the **electron detachment**. These processes of electron losses, related to negative ions will be considered in Section 2.4. Note that although electron losses in this second group are due to the electron attachment processes, the actual losses of charged particles take place as a consequence following the fast processes of **ion–ion recombination.** The ion–ion recombination process means neutralization during the collision of negative and positive ions. These processes usually have very high rate coefficients and will be considered in Section 2.5.

Finally, the third group of charged particle losses is not a volumetric one like all those mentioned above, but is due to **surface recombination**. These processes of electron losses are the most important in low-pressure plasma systems, like glow discharges. The surface recombination processes are usually kinetically limited not by the elementary act of the electron–ion recombination on the surface, but by transfer (diffusion) of the charged particles to the walls of the discharge chamber. For this reason, the surface losses of charged particles will be discussed later in Chapter 4, concerning plasma kinetics and transfer phenomena.

### 2.3.1 Different Mechanisms of Electron–Ion Recombination

The electron–ion recombination is a highly exothermic process like all other recombination processes. The released energy corresponds to the ionization potential and so it is quite high (see Table 2.1). To be effectively realized, such a process should obviously have a channel of accumulation of the energy released during the neutralization of a positive ion and an electron. Taking into account three main channels of consumption of the recombination energy listed below (dissociation, three-body collisions, and radiation), the electron–ion recombination processes can be subdivided into three principal groups of mechanisms. In molecular gases, or just in the presence of molecular ions, the fastest electron neutralization mechanism is **dissociative electron–ion recombination**:

$$e + AB^+ \rightarrow \left(AB\right)^* \rightarrow A + B^*. \tag{2.41}$$

In these processes, the recombination energy is usually going via resonance to dissociation of the molecular ion and to excitation of the dissociation products. While these processes principally take place in molecular gases, they can occur even in atomic gases by means of preliminary formation of molecular ions in the **ion conversion processes**: $A^+ + A + A \rightarrow A_2^+ + A$.

In atomic gases, in the absence of molecular ions, the neutralization can be due to **three-body electron–ion recombination:**

$$e + e + A^+ \rightarrow A^* + e. \tag{2.42}$$

The excessive energy in this case is going to the kinetic energy of a free electron, which participates in the recombination act as "a third body partner". Heavy particles (ions and neutrals) are unable to accumulate electron recombination energy fast enough in their kinetic energy and are ineffective as the third body partner. Finally, the recombination energy can be converted into radiation in the process of **radiative electron–ion recombination:**

$$e + A^+ \rightarrow A^* \rightarrow A + \hbar\omega \tag{2.43}$$

The cross section of this process is relatively low, and it can be competitive with the three-body recombination only when the plasma density is low.

### 2.3.2 The Dissociative Electron–Ion Recombination

This resonant recombination process Eq. (2.41) starts with the trapping of an electron on a repulsive auto-ionization level of molecular ion. Then atoms travel apart on the repulsive term, and if the auto-ionization state is maintained, stable products of dissociation are formed. This recombination mechanism is quite fast and plays a major role in the neutralization and decay of plasma in molecular gases. Reaction rate coefficients for most of the diatomic and three-atomic ions are on the level of $10^{-7}$ cm$^3$/s, for some important molecular ions the kinetic information can be found in Table 2.4.

In a group of similar ions, like molecular ions of noble gases, the recombination reaction rate is growing with several internal electrons: recombination of $Kr_2^+$ and $Xe_2^+$ is about 100 times faster than in case of helium. The rate coefficient of dissociative electron–ion recombination (the highly exothermic process) obviously decreases with temperature, as one can see that from Table 2.4. This process has no activation energy, so its dependencies on both electron $T_e$ and gas $T_0$ temperatures are not very strong. This dependence can be estimated (M. Biondi, 1976, V. Rusanov, A. Fridman, 1984) as:

$$k_r^{ei}\left(T_e, T_0\right) \propto \frac{1}{T_0\sqrt{T_e}}. \tag{2.44}$$

**TABLE 2.4**
**The Dissociative Electron–Ion Recombination Reaction Rate Coefficients at Room Gas Temperature $T_0$ = 300 K (Electron Temperature $T_e$ = 300 K and $T_e$ = 1 eV)**

| Electron–Ion Dissociative Recombination Process | Rate Coefficient $k_r^{ei}$, cm³/s $T_e = T_0$ = 300 K | Rate Coeff. $k_r^{ei}$, cm³/s $T_e$ = 1 eV, $T_0$ = 300 K | Electron–Ion Dissociative Recombination Process | Rate Coefficient $k_r^{ei}$, cm³/s $T_e = T_0$ = 300 K | Rate Coeff. $k_r^{ei}$, cm³/s $T_e$ = 1 eV, $T_0$ = 300 K |
|---|---|---|---|---|---|
| $e + N_2^+ \rightarrow N + N$ | $2*10^{-7}$ | $3*10^{-8}$ | $e + NO^+ \rightarrow N + O$ | $4*10^{-7}$ | $6*10^{-8}$ |
| $e+O_2^+ \rightarrow O+O$ | $2*10^{-7}$ | $3*10^{-8}$ | $e + H_2^+ \rightarrow H + H$ | $3*10^{-8}$ | $5*10^{-9}$ |
| $e+H_3^+ \rightarrow H_2+H$ | $2*10^{-7}$ | $3*10^{-8}$ | $e + CO^+ \rightarrow C + O$ | $5*10^{-7}$ | $8*10^{-8}$ |
| $e + CO_2^+ \rightarrow CO + O$ | $4*10^{-7}$ | $6*10^{-8}$ | $e + He_2^+ \rightarrow He + He$ | $10^{-8}$ | $2*10^{-9}$ |
| $e + Ne_2^+ \rightarrow Ne + Ne$ | $2*10^{-7}$ | $3*10^{-8}$ | $e + Ar_2^+ \rightarrow Ar + Ar$ | $7*10^{-7}$ | $10^{-7}$ |
| $e + Kr_2^+ \rightarrow Kr + Kr$ | $10^{-6}$ | $2*10^{-7}$ | $e + Xe_2^+ \rightarrow Xe + Xe$ | $10^{-6}$ | $2*10^{-7}$ |
| $e + N_4^+ \rightarrow N_2+N_2$ | $2*10^{-6}$ | $3*10^{-7}$ | $e + O_4^+ \rightarrow O_2 + O_2$ | $2*10^{-6}$ | $3*10^{-7}$ |
| $e + H_3O^+ \rightarrow H_2 + OH$ | $10^{-6}$ | $2*10^{-7}$ | $e + (NO)_2^+ \rightarrow 2NO$ | $2*10^{-6}$ | $3*10^{-7}$ |
| $e + HCO^+ \rightarrow CO + H$ | $2*10^{-7}$ | $3*10^{-8}$ | | | |

### 2.3.3 Ion Conversion Reactions as a Preliminary Stage of the Dissociative Electron–Ion Recombination

It is interesting, that if pressure is sufficiently high, the recombination of atomic ions like $Xe^+$ is increasing not by means of three-body Eq. (2.42) or radiative Eq. (2.43) mechanisms, but through the preliminary formation of molecular ions, through the so-called **ion conversion reactions** like $Xe^+ + Xe + Xe \rightarrow Xe_2^+ + Xe$. Then the molecular ion can be fast neutralized in the rapid process of dissociative recombination (see Table 2.4). The ion conversion reaction rate coefficients are quite high (B.M. Smirnov, 1974, L. Virin, V. Talrose, et al., 1978), see Table 2.5. When pressure exceeds 10 Torr, the ion conversion is usually faster than the following process of dissociative recombination, which becomes a limiting stage in the overall kinetics. An analytical expression for the ion-conversion three-body reaction rate coefficient can y derived from the dimension analysis (B.M. Smirnov, 1982):

$$k_{ic} \propto \left(\frac{\beta e^2}{4\pi\varepsilon_0}\right)^{5/4} \frac{1}{M^{1/2}T_0^{3/4}} \tag{2.45}$$

In this relation M and β are mass and polarization coefficient of colliding atoms, $T_0^-$ is the gas temperature. The ion conversion effect takes place as a preliminary stage of recombination not only for simple atomic ions, as it was discussed above, but also for some important molecular ions. As is clear from Table 2.4, the polyatomic ions have very high recombination rates, often exceeding $10^{-6}$ $cm^3/s$ at room temperature. This results in an interesting fact; the recombination of molecular ions like $N_2^+$, and $O_2^+$ at elevated pressures sometimes goes through the intermediate formation of dimers like- $N_4^+$, and $O_4^+$.

### 2.3.4 Three-Body Electron–Ion Recombination

This three-body recombination process $e + e + A^+ \rightarrow A^* + e$ is the most important one for high-density equilibrium plasma with temperature about 1 eV when the concentration of molecular ions is very low (because of thermal dissociation). The recombination process starts with a three-body capture of an electron by a positive ion and the formation of a very highly excited atom with a binding energy of about $T_e$. The initially formed highly excited atom then loses its energy in step-by-step deactivation through electron impacts. A final relaxation step from the lower excited state to the ground state has a relatively high value of energy transfer and it is usually due to radiative transition. The three-body electron–ion recombination process is a reverse one with respect to the stepwise ionization, considered in Section 2.2.4. For this reason, the reaction rate coefficient of this recombination process can be derived from Eq. (2.26) for the stepwise ionization rate coefficient $k_i^s$ and from the Saha equation for ionization/recombination balance and equilibrium electron density:

$$k_r^{eei} = k_i^s \frac{n_0}{n_e n_i} = k_i^s \frac{g_0}{g_e g_i}\left(\frac{2\pi\hbar}{mT_e}\right)^{3/2} \exp\left(\frac{I}{T_e}\right) \approx \frac{e^{10}}{\left(4\pi\varepsilon_0\right)^5 \sqrt{mT_e^9}}. \tag{2.46}$$

**TABLE 2.5**
**Ion Conversion Reaction Rate Coefficients at Room Temperature**

| Ion conversion process | Reaction rate coefficient | Ion conversion process | Reaction rate coefficient |
|---|---|---|---|
| $N_2^+ + N_2 + N_2 \rightarrow N_4^+ + N_2^+$ | $8*10^{-29}$ $cm^6/s$ | $O_2^+ + O_2 + O_2 \rightarrow O_4^+ + O_2^+$ | $3*10^{-30}$ $cm^6/s$ |
| $H^+ + H_2 + H_2 \rightarrow H_3^+ + H_2$ | $4*10^{-29}$ $cm^6/s$ | $Cs^+ + Cs + Cs \rightarrow Cs_2^+ + Cs$ | $1.5*10^{-29}$ $cm^6/s$ |
| $He^+ + He + He \rightarrow He_2^+ + H$ | $9*10^{-32}$ $cm^6/s$ | $Ne^+ + Ne + Ne \rightarrow Ne_2^+ + Ne$ | $6*10^{-32}$ $cm^6/s$ |
| $Ar^+ + Ar + Ar \rightarrow Ar_2^+ + Ar$ | $3*10^{-31}$ $cm^6/s$ | $Kr^+ + Kr + Kr \rightarrow Kr_2^+ + Kr$ | $2*10^{-31}$ $cm^6/s$ |
| $Xe^+ + Xe + Xe \rightarrow Xe_2^+ + Xe$ | $4*10^{-31}$ $cm^6/s$ | | |

Here $n_e$, $n_i$, $n_0$ are number densities of electrons, ions, and neutrals, $g_e$, $g_i$, $g_0$ are their statistical weights, e, m are electron charge and mass, electron temperature $T_e$ is taken as usually in energy units, I is an ionization potential. It is convenient for practical calculations to rewrite Eq. (2.46) as:

$$k_r^{eei}, \frac{cm^6}{\sec} = \frac{\sigma_0}{I} 10^{-14} \left( I \big/ T_e \right)^{4.5} \tag{2.47}$$

In this numerical formula: $\sigma_0$, $cm^2$, is the gas-kinetic cross section, the same as those involved in Eqs. (2.20) and (2.21), I and $T_e$ are the ionization potential and electron temperature, to be taken in eV.

The rate coefficient of the recombination process depends strongly on electron temperature but not exponentially. The typical value of $k_r^{eei}$ at room temperature is about $10^{-20}$ $cm^6/s$, at $T_e = 1$ eV this rate coefficient is about $10^{-27}$ $cm^6/s$. At room temperature, the three-body recombination is able to compete with dissociative recombination when electron concentration is quite high and exceeds $10^{13}$ $cm^{-3}$. If the electron temperature is about 1 eV, three-body recombination can compete with the dissociative recombination only in the case of exotically high electron density exceeding $10^{20}$ $cm^{-3}$. As was mentioned earlier, the excessive energy in the process under consideration is going to the kinetic energy of a free electron – the "third body". In this case, heavy particles ion and neutrals are too slow and ineffective as "third body partners. The correspondent rate coefficient of the third-order reaction is about $10^8$ times lower.

### 2.3.5 Radiative Electron–Ion Recombination

This electron–ion recombination process $e + A^+ \rightarrow A + \hbar\omega$ is a relatively slow one because it requires a photon emission during a short interval of the electron–ion interaction. This type of recombination can play a major role in the balance of charged particles only in the absence of molecular ions and in addition if the plasma density is quite low and the three-body mechanisms are suppressed. Typical values of cross sections of the radiative recombination process are about $10^{-21}$ $cm^2$, which is low. The reaction rate coefficients are therefore not high and can be simply estimated (Ya. B. Zeldovich, Yu. P. Raizer, 1966) as a function of electron temperature:

$$k_{rad.rec.}^{ei} \approx 3 \cdot 10^{-13} \left( T_e, eV \right)^{-3/4}, \mathrm{cm}^3 / \mathrm{s} \tag{2.48}$$

Comparing numerical relations (2.47) and (48), one can see, that the reaction rate of radiative recombination exceeds that of the three-body process when electron concentration is low:

$$n_e < 3 \cdot 10^{13} \left( T_e, eV \right)^{3.75}, \mathrm{cm}^3 / \mathrm{s}. \tag{2.49}$$

## 2.4 ELECTRON LOSSES DUE TO FORMATION OF NEGATIVE IONS: ELECTRON ATTACHMENT AND DETACHMENT PROCESSES

A simple illustration of the electron attachment process and negative ion formation was presented in Figure 2.1b. The electron balance in electro-negative gases, which have high EA (oxygen, different halogens, and their compounds, see Table 2.2) can be strongly affected by these processes. In some other gases, like $CO_2$ and $H_2O$, the electron attachment can be equally important, because of high EA of decomposition products and formation of negative ions like $O^-$ and $H^-$ during dissociation. The electron attachment processes are essential in the electron balance in weakly ionized plasma with low electron concentration and a low degree of ionization. This fact is due to the first kinetic order of this process with respect to the concentration of electrons, which means the reaction rate of the dissociative attachment is proportional to electron density. In contrast, the

recombination processes are of second or third kinetic order. This means the electron–ion recombination reaction rates are proportional to the square or cube of electron concentration, and therefore they can be neglected with respect to electron attachment if the electron density and ionization degree are low.

The negative ions formed as a result of electron attachment can be neutralized quite easy and extremely fast in the following processes of ion–ion recombination. For this reason, the attachment processes in general can provide significant losses of electrons and charged particles, which can result in the prevention of ignition and propagation of electric discharges. Fortunately for plasma generation, the electron attachment processes can be effectively suppressed by the reverse reactions of electron detachment. These processes of decay of negative ions and restoration of free electrons can be provided by collisions with excited or chemically active species, energetic electrons, etc.

### 2.4.1 Dissociative Electron Attachment to Molecules

It is important in molecular gases like $CO_2$, $H_2O$, $SF_6$, $CF_4$, when dissociation products have positive electron affinities, see (Table 2.2):

$$e + AB \rightarrow \left(AB^-\right)^* \rightarrow A + B^- \tag{2.50}$$

The mechanism of this process is somewhat similar to the dissociative recombination and proceeds by intermediate formation of an auto-ionization state $(AB^-)^*$. This excited state is unstable and its decay leads either to the reverse process of auto-detachment (AB + e) or to dissociation ($A + B^-$). During the attachment Eq. (2.50), an electron is captured and not able to provide an energy balance of the elementary process. For this reason, the dissociative attachment is a resonant reaction, which means it requires quite definite values of the electron energy. The most typical potential energy curves, illustrating the dissociative attachment Eq. (2.50), are presented in Figure 2.6a. The electron attachment process starts in this case with a vertical transition from AB molecular ground state electronic term to a repulsive state of $AB^-$ (obviously following the Frank-Condon Principle, Section 2.2.3). During the repulsion, before the $(AB^-)^*$ reaches an intersection point of AB and $AB^-$ electronic terms, the reverse auto-detachment reaction (AB + e) is very possible. But after passing the intersection, the AB potential energy exceeds that of $AB^-$ and further repulsion results in dissociation ($A + B^-$). To estimate the cross section of the dissociative attachment we should take into account, that the repulsion time with possible auto-detachment is proportional to the square root of reduced mass of AB molecule $\sqrt{M_A M_B / \left(M_A + M_B\right)}$ (H. Massey, 1976). The characteristic electron

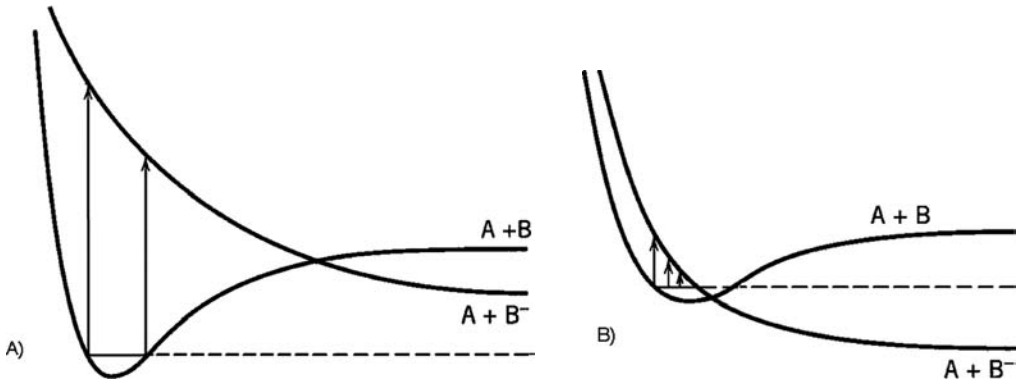



**FIGURE 2.6** Dissociative attachment: (a) low electron affinity, (b) high electron affinity.

transition time is much shorter and proportional to the square root of its mass m. For this reason (M.A. Lieberman, A.J. Lichtenberg, 1994), the maximum cross section of dissociative attachment with the described configuration of electronic terms can be estimated as:

$$\sigma_{d.a.}^{\max} \approx \sigma_0 \sqrt{\frac{m\left(M_A + M_B\right)}{M_A M_B}} \tag{2.51}$$

The maximum cross section is two orders of value less than the gas-kinetic cross section $\sigma_{0,}$ and numerically is of about $10^{-18}$ cm$^2$. Some change in the reaction cross section Eq. (2.51) occurs, if the $AB^-$ electronic term corresponds not to the repulsive, but rather to the attractive state (H. Massey, 1976),; this not a common situation. A more interesting process occurs when the EA of a product exceeds the dissociation energy. In particular, it takes place for some halogens and their compounds. Corresponding potential energy curves, illustrating the dissociative attachment, are presented in Figure 2.6b. In this case in contrast to the process depicted by Figure 2.6a, even very low energy electrons can effectively provide the dissociative attachment. Also, if the EA of a product exceeds the dissociation energy, the intersection point of AB and $AB^-$ electronic terms (Figure 2.6b) is actually located inside of the so-called geometrical sizes of the dissociating molecules. As a result, during the repulsion of $(AB^-)^*$, the probability of the reverse auto-detachment reaction (AB + e) is very low and cross section of the dissociative attachment process can in this case reach the gas-kinetic cross section $\sigma_0$ of about $10^{-16}$ cm$^2$.

Cross sections of the dissociative attachment for several molecular gases are presented in Figure 2.7 as a function of electron energy. Usually, only electrons, having enough energy (more than the difference between dissociation energy and EA), can provide the dissociative attachment. Also, the dissociative attachment cross section as a function of electron energy $\sigma_a(\varepsilon)$ has a resonant structure. It reflects the resonance nature of the process, which can be effective only within a narrow range of electron energy. The resonant structure of $\sigma_a(\varepsilon)$ permits estimating the dissociative attachment rate coefficient $k_a$ as a function of electron temperature $T_e$ as:

$$k_a\left(T_e\right) \approx \sigma_{d.a.}^{\max}\left(\varepsilon_{\max}\right)\sqrt{\frac{2\varepsilon_{\max}}{m}}\frac{\Delta\varepsilon}{T_e}\exp\left(-\frac{\varepsilon_{\max}}{T_e}\right) \tag{2.52}$$

In this relation the only resonance leading to the dissociative attachment and hence only the resonance of the $\sigma_a(\varepsilon)$ was taken into account, $\varepsilon_{max}$ and $\sigma^{max}_{d.a.}$ are the electron energy and maximum cross section (see Eq. (2.51)), corresponding to the resonance, $\Delta\varepsilon$ is its energy width. Equation (2.52) is quite convenient for numerical calculations; the necessary parameters: $\varepsilon_{max}$, $\sigma^{max}_{d.a.}$ and $\Delta\varepsilon$

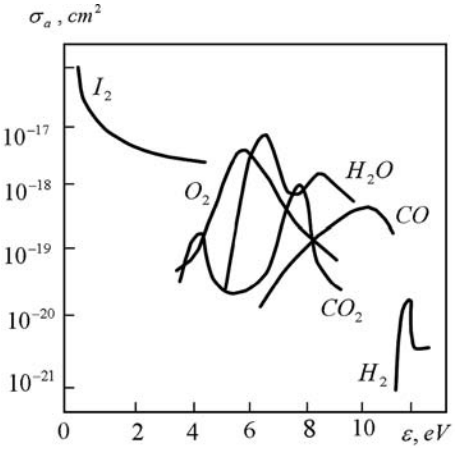



**FIGURE 2.7** Cross-sections of dissociative electron attachment to different molecules.

**TABLE 2.6**
**The Resonance Parameters for the Dissociative Attachment of Electrons to Different Molecules**

| Dissociative Attachment Process | $\varepsilon_{max}$, eV | $\sigma^{max}_{d.a}$, cm$^2$ | $\Delta\varepsilon$, eV |
|---|---|---|---|
| $e + O_2 \rightarrow O^- + O$ | 6.7 | $10^{-18}$ | 1 |
| $e + H_2 \rightarrow H^- + H$ | 3.8 | $10^{-21}$ | 3.6 |
| $e + NO \rightarrow O^- + N$ | 8.6 | $10^{-18}$ | 2.3 |
| $e + CO \rightarrow O^- + C$ | 10.3 | $2*10^{-19}$ | 1.4 |
| $e + HCl \rightarrow Cl^- + H$ | 0.8 | $7*10^{-18}$ | 0.3 |
| $e + H_2O \rightarrow H^- + OH$ | 6.5 | $7*10^{-18}$ | 1 |
| $e + H_2O \rightarrow O^- + H_2$ | 8.6 | $10^{-18}$ | 2.1 |
| $e + H_2O \rightarrow H + OH^-$ | 5 | $10^{-19}$ | 2 |
| $e + D_2O \rightarrow D^- + OD$ | 6.5 | $5*10^{-18}$ | 0.8 |
| $e + CO_2 \rightarrow O^- + CO$ | 4.35 | $2*10^{-19}$ | 0.8 |

are provided in Table 2.6. As was mentioned above, the information is given in each case only for one resonance, which is supposed to have the maximum contribution to kinetics of the dissociative attachment.

### 2.4.2 Three-Body Electron Attachment to Molecules

These attachment processes of formation of negative ions in a collision of an electron with two heavy particles (at least one of them is supposed to have positive EA) can be presented as:

$$e + A + B \rightarrow A^- + B \tag{2.53}$$

The three-body electron attachment can be a principal channel of electron losses through formation of negative ions when electron energies are not high enough for the dissociative attachment, and when pressure is elevated (usually more than 0.1 atm) and the third kinetic order processes are preferable. In contrast to the dissociative attachment, the three-body process doesn't require the consumption of electron energy. For this reason, its rate coefficient does not depend strongly on electron temperature (at least, within the temperature range about 1 eV). In this case, heavy particles are responsible for the consumption of energy released during the attachment. Electrons are usually kinetically not effective as a third body B, because of low degree of ionization and low energy release during the attachment (in contrast to the three-body electron–ion recombination Eq. (2.42)).

Atmospheric pressure nonthermal discharges in air are systems, where the three-body attachment plays a key role in the balance of charged particles. That is why the three-body electron attachment to molecular oxygen ($e + O_2 + M \rightarrow O_2^- + M$) can be taken as a good example. The three-body electron attachment to molecular oxygen proceeds by the two-stage **Bloch-Bradbury mechanism** (F. Bloch, N. Bradbury, 1935; N. Alexandrov, 1981). The first stage of the process includes an electron attachment to the molecule with the formation of a negative oxygen ion in an unstable autoionization state:

$$e + O_2 \overset{k_{att},\tau}{\leftrightarrow} \left(O_2^-\right)^*. \tag{2.54}$$

Here $k_{att}$ is a rate coefficient of the intermediate electron trapping, $\tau$ is lifetime of the excited unstable ion with respect to collisionless decay into the initial state. The second stage of the Bloch-Bradbury mechanism includes collision with the third-body particle M with either relaxation and

stabilization of $O_2^-$ (rate coefficient $k_{st}$) or collisional decay of the unstable ion into initial state (rate coefficient $k_{dec}$):

$$\left(O_2^-\right)^* + M \xrightarrow{k_{st}} O_2^- + M. \tag{2.55}$$

$$\left(O_2^-\right)^* + M \xrightarrow{k_{dec}} O_2 + e + M. \tag{2.56}$$

Taking into account the steady-state conditions for number density of the intermediate excited ions $(O_2^-)^*$, the rate coefficient for the total process of three-body electron attachment to molecular oxygen $e + O_2 + M \rightarrow O_2^- + M$ can be expressed in the following way:

$$k_{3M} = \frac{k_{att}k_{st}}{\frac{1}{\tau} + \left(k_{st} + k_{dec}\right)n_0}. \tag{2.57}$$

In this relation, $n_0$ is the concentration of the third-body heavy particles M When the pressure is not too high, $(k_{st} + k_{dec})n_0 << \tau^{-1}$, and Eq. (2.57) for the reaction rate can be significantly simplified:

$$k_{3M} \approx k_{att}k_{st}\tau \tag{2.58}$$

The three-body attachment process has a third kinetic order. It equally depends on the rate coefficient of formation and stabilization of negative ions on a third particle. The latter one strongly depends on the type of the third particle. In general, the more complicated molecule plays a role of the third body (M), the easier it stabilizes the $(O_2^-)^*$ and the higher is the total reaction rate coefficient $k_{3M.}$ Numerical values of the total rate coefficients $k_{3M}$ are presented in Table 2.7 for room temperature:

The three-body attachment rate coefficients are shown on the Figure 2.8 as a function of electron temperature (the gas, in general, is taken to be room temperature). For simple estimations,

**TABLE 2.7**
**Reaction Rate Coefficients of the Electron Attachment to Oxygen Molecules at Room Temperature with Different Third Body Partners**

| Three-Body Attachment | Rate coefficient | Three-Body Attachment | Rate coefficient |
|---|---|---|---|
| $e + O_2 + Ar \rightarrow O_2^- + Ar$ | $3*10^{-32}$ cm$^6$/s | $e + O_2 + Ne \rightarrow O_2^- + Ne$ | $3*10^{-32}$ cm$^6$/s |
| $e + O_2 + N_2 \rightarrow O_2^- + N_2$ | $1.6*10^{-31}$ cm$^6$/s | $e + O_2 + H_2 \rightarrow O_2^- + H_2$ | $2*10^{-31}$ cm$^6$/s |
| $e + O_2 + O_2 \rightarrow O_2^- + O_2$ | $2.5*10^{-30}$ cm$^6$/s | $e + O_2 + CO_2 \rightarrow O_2^- + CO_2$ | $3*10^{-30}$ cm$^6$/s |
| $e + O_2 + H_2O \rightarrow O_2^- + H_2O$ | $1.4*10^{-29}$ cm$^6$/s | $e + O_2 + H_2S \rightarrow O_2^- + H_2S$ | $10^{-29}$ cm$^6$/s |
| $e + O_2 + NH_3 \rightarrow O_2^- + NH_3$ | $10^{-29}$ cm$^6$/s | $e + O_2 + CH_4 \rightarrow O_2^- + CH_4$ | $>10^{-29}$ cm$^6$/s |

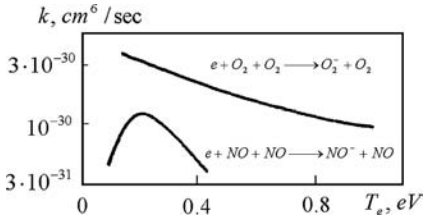



**FIGURE 2.8** Rate coefficients of the three body electron attachment to molecules.

when $T_e$ = 1 eV, $T_0$ = 300 K, one can take $k_{3M} \approx 10^{-30}$ cm$^6$/s. The rate of the three-body process is greater than dissociative attachment ($k_a$) when the gas number density exceeds a critical value $n_0 > k_a(T_e) / k_{3M}$. Numerically in oxygen with $T_e$ = 1 eV, $T_0$ = 300 K, this means $n_0 > 10^{18}$ cm$^{-3}$, or in pressure units p > 30 Torr.

### 2.4.3 Other Mechanisms of Formation of Negative Ions

Let us just mention three other mechanisms of formation of negative ions, which can be usually neglected, but in some specific situations should be taken into account. The first process is **the polar dissociation**:

$$e + AB \rightarrow A^+ + B^- + e. \tag{2.59}$$

This process actually includes both ionization and dissociation, therefore the threshold energy is quite high in this case. On the other hand, the electron is not captured, the process is not a resonant one and can be effective in a wide range of high electron energies. For example in molecular oxygen, the maximum value of cross section of the polar dissociation is about $3*10^{-19}$ cm$^2$ and corresponds to electron energy 35 eV. To stabilize the formation of a negative ion during electron attachment, the excessive energy of an intermediate excited ion particle can be emitted. As a result, a negative ion can be formed in the process of radiative attachment:

$$e + M \rightarrow \left(M^-\right)^* \rightarrow M^- + \hbar\omega. \tag{2.60}$$

Such an electron capture can take place at low electron energies, but the probability of the radiative process is very low, about $10^{-5}$–$10^{-7}$. Corresponding values of attachment cross sections are about $10^{-21}$–$10^{-23}$ cm$^2$. Finally, some electronegative polyatomic molecules like $SF_6$ have a negative ion state very close to a ground state (only 0.1 eV in the case of $SF_6^-$). As a result, the lifetime of such metastable negative ions can be long. Such an attachment process is resonant and for very low electron energies has maximum cross sections of about $10^{-15}$ cm$^2$.

### 2.4.4 Mechanisms of Negative Ion Destruction, Associative Detachment Processes

The negative ions can be then neutralized in the fast ion–ion recombination or can release an electron by its detachment and ion destruction. Competition between these two processes defines the balance of charged particles and often determines regime of the discharges. Different mechanisms can provide the destruction of negative ions with an electron release. These mechanisms are discussed in detail in special books and reviews: H. Massey, 1976; E.W. McDaniel, 1964; B.M. Smirnov, 1978; N. Alexandrov, 1981. Let us consider three detachment mechanisms, which are most important in plasma-chemical systems. In nonthermal discharges, probably the most important one is the **associative detachment**:

$$A^- + B \rightarrow \left(AB^-\right)^* \rightarrow AB + e. \tag{2.61}$$

This process is a reverse one with respect to the dissociative attachment Eq. (2.50) and so it can also be illustrated by the same Figure 2.6. The associative detachment is a non-adiabatic process, which occurs by the intersection of electronic terms of a complex negative ion $A^-$ – B and corresponding molecule AB. For this reason (non-adiabatic reaction), the rate coefficients of such processes are usually quite high. Typical values of the coefficients are about $k_d = 10^{-10}$–$10^{-9}$ cm$^3$/s, and are not far from those of gas-kinetic collisions. Because the reaction (2.61) corresponds to the intersection of electronic terms of $A^-$– B and AB, it can have an energy barrier even in exothermic

**TABLE 2.8**
**The Associative Attachment Rate Coefficients at Room Temperature**

| Associative Attachment Process | Reaction Enthalpy | Rate Coefficient |
|---|---|---|
| $H^- + H \rightarrow H_2 + e$ | −3.8 eV (exothermic) | $1.3*10^{-9}$ cm$^3$/s |
| $H^- + O_2 \rightarrow HO_2 + e$ | −1.25 eV (exothermic) | $1.2*10^{-9}$ cm$^3$/s |
| $O^- + O \rightarrow O_2 + e$ | −3.8 eV (exothermic) | $1.3*10^{-9}$ cm$^3$/s |
| $O^- + N \rightarrow NO + e$ | −5.1 eV (exothermic) | $2*10^{-10}$ cm$^3$/s |
| $O^- + O_2 \rightarrow O_3 + e$ | 0.4 eV (endothermic) | $10^{-12}$ cm$^3$/s |
| $O^- + O_2(^1\Delta_g) \rightarrow O_3 + e$ | −0.6 eV (exothermic) | $3*10^{-10}$ cm$^3$/s |
| $O^- + N_2 \rightarrow N_2O + e$ | −0.15 eV (exothermic) | $10^{-11}$ cm$^3$/s |
| $O^- + NO \rightarrow NO_2 + e$ | −1.6 eV (exothermic) | $5*10^{-10}$ cm$^3$/s |
| $O^- + CO \rightarrow CO_2 + e$ | −4 eV (exothermic) | $5*10^{-10}$ cm$^3$/s |
| $O^- + H_2 \rightarrow H_2O + e$ | −3.5 eV (exothermic) | $10^{-9}$ cm$^3$/s |
| $O^- + CO_2 \rightarrow CO_3 + e$ | (endothermic) | $10^{-13}$ cm$^3$/s |
| $C^- + CO_2 \rightarrow CO + CO + e$ | −4.3 (exothermic) | $5*10^{-11}$ cm$^3$/s |
| $C^- + CO \rightarrow C_2O + e$ | −1.1 (exothermic) | $4*10^{-10}$ cm$^3$/s |
| $Cl^- + O \rightarrow ClO + e$ | 0.9 (endothermic) | $10^{-11}$ cm$^3$/s |
| $O_2^- + O \rightarrow O_3 + e$ | −0.6 eV (exothermic) | $3*10^{-10}$ cm$^3$/s |
| $O_2^- + N \rightarrow NO_2 + e$ | −4.1 eV (exothermic) | $5*10^{-10}$ cm$^3$/s |
| $OH^- + O \rightarrow HO_2 + e$ | −1 eV (exothermic) | $2*10^{-10}$ cm$^3$/s |
| $OH^- + N \rightarrow HNO + e$ | −2.4 eV (exothermic) | $10^{-11}$ cm$^3$/s |
| $OH^- + H \rightarrow H_2O + e$ | −3.2 eV (exothermic) | $10^{-9}$ cm$^3$/s |

conditions. Rate coefficients of some associative attachment processes are presented in Table 2.8 together with an enthalpy of the reactions.

The associative attachment is quite fast. For example, the reaction $O^- + CO \rightarrow CO_2 + e$ can effectively suppress the dissociative attachment in $CO_2$ and corespondent electron losses, if CO concentration is high enough. This effect is of importance in nonthermal discharges in $CO_2$ and in $CO_2^-$ laser mixture. We should note though, that fast three-body cluster formation processes are able to stabilize $O^-$ with respect to associative attachment. The following clusterization reactions:

$$O^- + CO_2 + M \rightarrow CO_3^- + M, k = 10^{-27}\,\text{cm}^6/\text{s} \tag{2.62}$$

$$O^- + O_2 + M \rightarrow O_3^- + M, k = 10^{-30}\,\text{cm}^6/\text{sec} \tag{2.63}$$

converts $O^-$ ions into $CO_3^-$, which are more stable with respect to detachment and hence promote ion–ion recombination and loss of charged particles.

### 2.4.5 Electron Impact Detachment

This detachment process can be described as follows: $e + A^- \rightarrow A + 2e$ and is an essential one in the balance of negative ions when the ionization degree is high. The electron impact detachment is somewhat similar to the direct ionization of neutrals by an electron impact (Thomson mechanism, see section 2.2.1). The main difference in this case is due to repulsive Coulomb force acting between the incident electron and the negative ion. For incident electron energies about 10 eV, the cross section of the detachment process can be high, about $10^{-14}$ cm$^2$. Typical detachment cross section dependence on energy or on the incident electron velocity $v_e$ (M. Inokuti, Y. Kim, R.L. Platzman, 1967) can be illustrated by that for the detachment from negative hydrogen ion ($e + H^- \rightarrow H + 2e$):

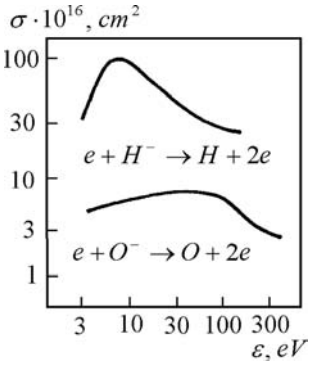



**FIGURE 2.9** Negative ion destruction by an electron impact.

$$\sigma\left(v_e\right) \approx \frac{\sigma_0 e^4}{\left(4\pi\varepsilon_0\right)^2 \hbar^2 v_e^2}\left(-7.5\ln\frac{e^2}{4\pi\varepsilon_0\hbar v_e} + 25\right) \tag{2.64}$$

In this relation, $\sigma_0$ is the geometrical atomic cross section, more exactly defined in Eq. (2.20). The detachment cross section as a function of electron energy is presented in Figure 2.9 for hydrogen and oxygen atomic ions. The maximum cross sections of about $10^{-15}$–$10^{-14}$ cm$^2$ corresponds to electron energies about 10–50 eV. These energies exceed electron affinities more than 10 times in contrast to the Thomson mechanism of the electron impact ionization (which can be explained by the Coulomb repulsion).

### 2.4.6 Detachment in Collisions with Excited Particles

This detachment process ($A^- + B^* \rightarrow A + B + e$), for the case of using **electronic excitation** energy of particle B, is similar to the Penning ionization (see section 2.2.8). If the electronic excitation energy of a collision partner B exceeds the EA of another particle - A, the detachment process can proceed effectively as an electronically non-adiabatic reaction (without significant energy exchange with translational degrees of freedom of the heavy particles, see section 2.2.7). As an example, consider the exothermic detachment of an electron from an oxygen ion in collision with a metastable electronically excited oxygen (excitation energy 0.98 eV):

$$O_2^- + O_2\left(^1\Delta_g\right) \rightarrow O_2 + O_2 + e, \Delta H = -0.6 \text{ eV}. \tag{2.65}$$

The rate coefficient of the detachment reaction is very high, even at room temperature it is about $2*10^{-10}$ cm$^3$/s. Electron detachment also can be effective in collisions with **vibrationally excited molecules**. Consider again as an example the destruction of an oxygen molecular ion: $O_2^- + O_2^*(v > 3) - O_2 + O_2 + e$. Rate coefficients of the detachment process in the quasi-equilibrium systems (characterized by a single temperature) are presented in the following Table 2.9. The first process

**TABLE 2.9**
**Thermal Destruction of an Oxygen Molecular Ion**

| Detachment Process | Reaction enthalpy | Rate Coefficient, 300K | Rate Coefficient, 600K |
|---|---|---|---|
| $O_2^- + O_2 \rightarrow O_2 + O_2 + e$ | 0.44 eV | $2.2*10^{-18}$ cm$^3$/s | $3*10^{-14}$ cm$^3$/s |
| $O_2^- + N_2 \rightarrow O_2 + N_2 + e$ | 0.44 eV | $<10^{-20}$ cm$^3$/s | $1.8*10^{-16}$ cm$^3$/s |

presented in the Table has the activation energy close to the reaction enthalpy. These quasi-equilibrium detachment processes are using mostly vibrational energy of colliding partners, because in other cases the reaction is strongly adiabatic (see Section 2.2.7). The detachment process stimulated by vibrational excitation of molecules proceeds according to the modified Bloch-Bradbury mechanism Eqs. (2.53–2.55). In this case, the process starts with collisional excitation of $O_2^-$ to the vibrationally excited states with v > 3. The excited ions $(O_2^-)^*(v > 3)$ are then in the state of auto-ionization, which results in an electron detachment during a period of time shorter than the interval between two collisions. Excitation of $(O_2^-)^*(v > 3)$ is easier and faster in oxygen, than in nitrogen, where the process is less resonant. It explains the significant difference in detachment rate coefficients in collisions with oxygen and nitrogen, which one can see from Table 2.9. Kinetics of the detachment process can be described in this case in a conventional way for all reactions stimulated by vibrational excitation of molecules (see Chapter 5). The traditional Arrhenius formula: $k_d \propto \exp(-E_a/T_v)$ is applicable here, if molecular vibrations are in internal quasi-equilibrium with the only vibrational temperature $T_v$. The activation energy of the detachment process can be taken in this case equal to the EA to oxygen molecules ($E_a \approx 0.44$eV) in the same way one can describe detachment kinetics of $NO^-$, $F^-$ and $Br^-$ (H. Massey, 1976).

## 2.5 THE ION–ION RECOMBINATION PROCESSES

In electro-negative gases, when the attachment processes are involved in the balance of electrons and ions, the actual losses of charged particles are mostly due to ion–ion recombination. This process means mutual neutralization of positive and negative ions in binary or three-body collisions. The ion–ion recombination can proceed by a variety of different mechanisms, which dominate in different pressure ranges. However, all of them are characterized by very high rate coefficients. At high pressures (usually more than 30 Torr), three-body mechanisms dominate the recombination. The recombination rate coefficient in this case reaches a maximum of about $1–3*10^{-6}$ cm$^3$/s near atmospheric pressures for room temperature. Traditionally the rate coefficient of ion–ion recombination is recalculated with respect to concentrations of positive and negative ions, which is to the second kinetic order. Due to the three-molecular nature of the recombination mechanism in this pressure range, the recalculated (second kinetic order) recombination rate coefficient depends on pressure. Near atmospheric pressure corresponds to the fastest neutralization. Both an increase and a decrease of pressure results in proportional reduction of the three-body ion–ion recombination rate coefficient. At low pressures, the three-body mechanism becomes relatively slow. In this pressure range, the binary collisions with transfer of energy into electronic excitation make a major contribution to the ion–ion recombination.

### 2.5.1 Ion–Ion Recombination in Binary Collisions

Such neutralization proceeds in a collision of a negative and positive ion with the released energy going to electronic excitation of a neutral product:

$$A^- + B^+ \rightarrow A + B^* \tag{2.66}$$

This reaction has a second kinetic order and dominates the ion–ion recombination in low-pressure discharges (p < 10–30 Torr). Electronic terms, illustrating the recombination, are presented in Figure 2.10. The ions $A^-$ and $B^+$ are approaching each other following the attractive $A^- – B^+$ Coulomb potential curve. When the distance between the heavy particles is large, this potential curve lies below the term of $A^- – B^+$ and above the final $A – B^*$ electronic term on the energy interval:

$$\Delta E \approx I_B/\eta - EA_A \quad \Delta E \approx I_B/\eta \tag{2.67}$$

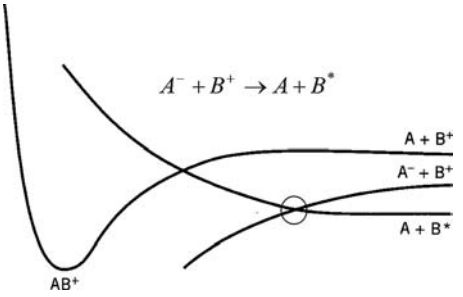



**FIGURE 2.10** Terms illustrating the ion-ion recombination.

Here $I_B$ is the ionization potential of particle B. If the principal quantum number of particle B after recombination is high n = 3–4, then ΔE is low. This means that the electronic terms of $A^- - B^+$ and A – B* are relatively close when the ions $A^-$ and $B^+$ are approaching each other. When the principal quantum number n is not specifically defined, then the affinity $EA_A$ can be taken as a reasonable estimation for the energy interval Eq. (2.67), which is $\Delta E \approx EA_A$. As can be seen from Figure 2.11, the low value of Δ Eresults in the possibility of effective transition between the electronic terms (from $A^- - B^+$ to A – B*), when distance $R_{ii}$ between ions is still large. Even for this long-distance $R_{ii}$ between the ions the Coulomb attraction energy is already sufficient to compensate the initial energy gap ΔE between the terms:

$$R_{ii} \approx \frac{e^2}{4\pi\varepsilon_0} \frac{1}{\left( I_B \big/ \eta^2 - \mathrm{EA}_A \right)} \approx \frac{e^2}{4\pi\varepsilon_0 \mathrm{EA}_A} \tag{2.68}$$

The high value of $R_{ii}$ results in large cross sections of the ion–ion recombination Eq. (2.66). This can be estimated from the conservation of angular momentum during the attractive Coulomb collision and assuming maximum kinetic energy equal to the potential energy $EA_A$ Eq. (4.5.2). The impact parameter b (see Section 2.1.7) can be found as a function of the ion kinetic energy $\varepsilon$ in the center of mass system:

$$b \approx R_{ii} \frac{\sqrt{\mathrm{EA}_A}}{\sqrt{\varepsilon}} \tag{2.69}$$

Based on the expressions for the impact parameter b and for the reactive distance $R_{ii}$, the formula to calculate the cross section of the ion–ion recombination process can be given as:

$$\sigma^{ii}_{rec} = \pi b^2 \approx \pi \frac{e^4}{\left(4\pi\varepsilon_0\right)^2} \frac{1}{\mathrm{EA}_A} \frac{1}{\varepsilon} \tag{2.70}$$

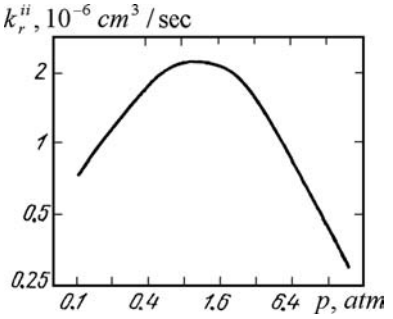



**FIGURE 2.11** Ion-ion recombination rate coefficient in air.

Quantum mechanical calculation of the ion–ion recombination across the section should take into account the effect of electron tunneling (B.M. Smirnov, 1978). As a result, it gives the more precise dependence of cross section of the ion–ion recombination in binary collisions of the ion kinetic energy as: $\sigma_{rec}^{ii} \propto 1/\sqrt{\varepsilon}$. Equation (2.70) ca be rewritten referring to the atomic units, $\sigma_{reci}^{ii} \approx \sigma_0 \frac{1}{\mathrm{EA}_A} \frac{1}{\varepsilon}$, where I is a typical value of ionization energy. This means that the recombination cross section exceeds the gas-kinetic one by several orders of magnitude. The same is true for the reaction rate coefficients, which usually are of the order of $10^{-7}$ cm³/s. Some binary recombination rate coefficients are presented in Table 2.10 together with the energy released in the process.

### 2.5.2 The Three-Body Ion–Ion Recombination, Thomson's Theory

The neutralization process can effectively proceed in a triple collision of a heavy neutral with a negative and a positive ion:

$$A^- + B^+ + M \rightarrow A + B + M \tag{2.71}$$

This process dominates the ion–ion recombination at moderate and high pressures (p > 10–30 Torr). The three-body reaction has the third kinetic order only in the moderate pressure range, usually less than 1 atm (for high pressures the process is limited by ion mobility, see below). It means that only at moderate pressures, the reaction rate proportional to the product of concentrations of all three collision partners, positive and negative ions, and neutrals. At pressures about (0.01 – 1 atm), the three-body ion–ion recombination can be described in the framework of the Thomson theory (Thomson, 1924). According to this theory, the act recombination takes place if a negative and positive ions approach each other closer than the critical distance $b \approx e^2/4\pi\varepsilon_0 T_0$. In this case, their Coulomb interaction energy reaches the level of thermal energy, which is approximately $T_0$ the gas temperature or temperature of heavy particles. Then during the collision, the third body (a heavy neutral particle) can absorb the energy of approximately equal to the thermal energy $T_0$ from an ion, and provide the act of recombination. The collision frequency of an ion with a neutral particle leading to energy transfer of about $T_0$ between them and subsequently resulting in recombination, can be found as $n_0 \sigma \nu_t$, where $n_0$ is the neutrals number density, $\sigma$ is a typical cross section of ion-neutral elastic scattering, $v_t$ is an average thermal velocity of the heavy particles. After the collision, the

**TABLE 2.10**
**Reaction Rate Coefficients of the Ion–Ion Recombination in Binary Collisions at Room Temperature**

| Recombination Process | Released Energy, I – EA | Rate Coefficient |
|---|---|---|
| $H^- + H^+ \rightarrow H + H$ | 12.8 eV | $3.9*10^{-7}$ cm³/s |
| $O^- + O^+ \rightarrow O + O$ | 12.1 eV | $2.7*10^{-7}$ cm³/s |
| $O^- + N^+ \rightarrow O + N$ | 13.1 eV | $2.6*10^{-7}$ cm³/s |
| $O^- + O_2^+ \rightarrow O + O_2$ | 11.6 eV | $10^{-7}$ cm³/s |
| $O^- + NO^+ \rightarrow O + NO$ | 7.8 eV | $4.9*10^{-7}$ cm³/s |
| $SF_6 + SF_5^+ \rightarrow SF_6 + SF_5$ | 15.2 eV | $4*10^{-8}$ cm³/s |
| $NO_2^- + NO^+ \rightarrow NO_2 + NO$ | 5.7 eV | $3*10^{-7}$ cm³/s |
| $O_2^- + O^+ \rightarrow O_2 + O$ | 13.2 eV | $2*10^{-7}$ cm³/s |
| $O_2^- + O_2^+ \rightarrow O_2 + O_2$ | 11.6 eV | $4.2*10^{-7}$ cm³/s |
| $O_2^- + N_2^+ \rightarrow O_2 + N_2$ | 15.1 eV | $1.6*10^{-7}$ cm³/s |

probability $P_+$ for a positive ion to be closer than $b \approx e^2/4\pi\varepsilon_0 T_0$ to a negative ion, and as a result to be neutralized in the following recombination can be estimated as:

$$P_+ \approx n_- b^3 \approx n_- \frac{e^6}{\left(4\pi\varepsilon_0\right)^3 T_0^3} \tag{2.72}$$

The frequency of the collisions of a positive ion with neutral particles, resulting in ion–ion recombination, can be then be found as $\nu_{ii} = (\sigma v_t n_0)P_+$. Consequently, the total three-body ion–ion recombination rate can be presented according to the Thomson's theory as:

$$w_{ii} \approx \left(\sigma v_t\right) \frac{e^6}{\left(4\pi\varepsilon_0\right)^3 T_0^3} n_0 n_- n_+ \tag{2.73}$$

The process has a third kinetic order with the rate coefficient $k_{r3}^{ii}$ decreasing with temperature:

$$k_{r3}^{ii} \approx \left(\sigma\sqrt{\frac{1}{m}}\right) \frac{e^6}{\left(4\pi\varepsilon_0\right)^3} \frac{1}{T_0^{5/2}} \tag{2.74}$$

At room temperature, the recombination coefficients are about $10^{-25}$ cm$^6$/s. For some reactions, these third-order kinetic reaction rate coefficients are presented in Table 2.11. The coefficients in the table are also recalculated to the second kinetic order $k_{r2}^{ii}$, which means $k_{r2}^{ii} = k_{r3}^{ii} n_0$. Comparing Tables 2.10 and 2.11, the binary and triple collisions are seen to contribute equally to the ion–ion recombination at pressures about 10–30 Torr.

### 2.5.3 High-Pressure Limit of the Three-Body Ion–Ion Recombination, Langevin Model

According to Eqs. (2.73) and (2.74), the ion–ion recombination rate coefficient recalculated to the second kinetic order $k_{r2}^{ii} = k_{r3}^{ii} n_0$ grows linearly with pressure at fixed temperatures. This growth is limited in the range of moderate pressures (usually not more than 1 atm) by the frameworks of Thomson's theory. Thomson's approach requires the capture distance b to be less than an ion mean free path ($1/n_0\sigma$). This requirement actually prevents using Thomson's theory for a high concentration of neutrals and high pressures (more than 1 atm):

$$n_0 \sigma b \approx n_0 \frac{e^2}{\left(4\pi\varepsilon_0\right) T_0} \sigma < 1 \tag{2.75}$$

**TABLE 2.11**
**Rate Coefficients of the Three-Body Ion–Ion Recombination Processes at Room Temperature and Moderate Pressures (In the Last Column the Coefficients Are Recalculated to the Second Kinetic Order and Pressure of 1 Atm, that Is Multiplied by the Concentration of Neutrals 2.7*10$^{19}$ cm$^{-3}$)**

| Ion–Ion Recombination | Rate Coefficients, 3rd order | Rate Coefficients, 2-rd order, 1atm |
|---|---|---|
| $O_2^- + O_4^+ + O \rightarrow O_2 + O_2 + O_2 + O_2 + O_2$ | 1.55*10$^{-25}$ cm$^6$/s | 4.2*10$^{-6}$ cm$^3$/s |
| $O^- + O_2^+ + O_2 \rightarrow O_3 + O_2$ | 3.7*10$^{-25}$ cm$^6$/s | 10$^{-5}$ cm$^6$/s |
| $NO_2^- + NO^+ + O_2 \rightarrow NO_2 + NO + O_2$ | 3.4*10$^{-26}$ cm$^6$/s | 0.9*10$^{-6}$ cm$^6$/s |
| $NO_2^- + NO^+ + N_2 \rightarrow NO_2 + NO + N_2$ | 10$^{-25}$ cm$^6$/s | 2.7*10$^{-6}$ cm$^6$/s |

In the opposite case of high pressures, $n_0\sigma b > 1$, the recombination is limited by the phase, in which a positive and negative ion approach each other, moving in electric field overcoming multiple collisions with neutrals. The Langevin model developed in 1903, describes the ion–ion recombination in this pressure range. This motion of a positive and negative ion to collide in recombination can be considered as their drift in the Coulomb field $e/(4\pi\varepsilon_0)r^2$, where r is the distance between them. Use positive and negative ion mobility $\mu_+$ and $\mu_-$ (see Chapter 4) as a coefficient of proportionality between ion drift velocity and the strength of the electric field. Then the ion drift velocity to meet each other is:

$$v_d = \frac{e}{\left(4\pi\varepsilon_0\right)^2}\left(\mu_+ + \mu_-\right). \tag{2.76}$$

Consider sphere with radius r, surrounding positive ion. A flux of negative ions with concentration $n_-$ approaching the positive one and the recombination frequency $\nu_{r+}$ for the positive ion are as follows:

$$\nu_{r+} = 4\pi r^2 v_d n_- \tag{2.77}$$

Based on Eq. (2.77) for neutralization frequency with respect to one positive ion, the total ion–ion recombination reaction rate can be found as: $w = n_+\nu_{r+} = 4\pi r^2 v_d n_- n_+$ . In this case, it shows the second kinetic order of the process with respect to ion concentrations. The final Langevin expression for the ion–ion recombination rate coefficient $k_r^{ii} = w / n_+ n_-$ in the limit of high pressures (usually more than 1 atm) can then be given taking into account Eqs. (2.76) and (2.77), as:

$$k_r^{ii} = 4\pi e\left(\mu_+ + \mu_-\right) \tag{2.78}$$

This recombination rate coefficient is proportional to the ion mobility, which decreases proportionally with pressure (see Chapter 4). Comparing the Thomson and Langevin models, it is seen that at first, recombination rate coefficients grow with pressure $k_{r2}^{ii} = k_{r3}^{ii} n_0$ (see Eq. (2.74)), and then at high pressures they begin to decrease as 1/p together with ion mobility Eq. (2.78). The highest recombination coefficient according to Eq. (2.75) corresponds to the concentration of neutrals:

$$n_0 \approx \frac{4\pi\varepsilon_0 T_0}{\sigma e^2} \tag{2.79}$$

At room temperature, this gas concentration corresponds to atmospheric pressure. The maximum value of the ion–ion recombination rate coefficient can be then found by substituting the number density $n_0$ into $k_{r2}^{ii} = k_{r3}^{ii} n_0$:

$$k_{r,\max}^{ii} \approx \frac{e^4}{\left(4\pi\varepsilon_0\right)^2}\frac{\nu}{T_0^2} \tag{2.80}$$

Numerically, the maximum recombination rate coefficient is about 1–3*10$^{-6}$ cm$^3$/s. This coefficient decreases for both an increase or decrease of pressure from atmospheric pressure. The total experimental dependence of the $k_r^{ii}$ on pressure is presented in Figure 2.11 The generalized ion–ion recombination model, combining the Thomson's and Langevin approaches for moderate and high pressures, was developed by Natanson (1959).

## 2.6 THE ION–MOLECULAR REACTIONS

### 2.6.1 Ion–Molecular Polarization Collisions, the Langevin Rate Coefficient

The ion–molecular processes can start with scattering in the polarization potential, leading to the so-called Langevin capture of a charged particle and formation of an intermediate ion-molecular complex. If a neutral particle itself has no permanent dipole moment, the ion-neutral charge–dipole

**TABLE 2.12**
**Polarizabilities of Atoms and Molecules**

| Atom or Molecule | A, $10^{-24}$ cm$^3$ | Atom or Molecule | A, $10^{-24}$ cm$^3$ |
|---|---|---|---|
| Ar | 1.64 | H | 0.67 |
| C | 1.78 | N | 1.11 |
| O | 0.8 | CO | 1.95 |
| $Cl_2$ | 4.59 | $O_2$ | 1.57 |
| $CCl_4$ | 10.2 | $CF_4$ | 1.33 |
| $H_2O$ | 1.45 | $CO_2$ | 2.59 |
| $SF_6$ | 4.44 | $NH_3$ | 2.19 |

interaction and scattering are due to the dipole moment pM, induced in the neutral particle by the electric field E of an ion:

$$p_m = \alpha\varepsilon_0 E = \alpha\frac{e}{4\pi r^2} \tag{2.81}$$

Here r is the distance between the interacting particles, $\alpha$ is the polarizability of a neutral atom or molecule, (see Table 2.12). The typical orbits of relative ion and neutral motion during polarization scattering are shown in Figure 2.12. When the impact parameter is high, the orbit has a hyperbolic character. But when the impact parameter is sufficiently low, the scattering leads to the Langevin polarization capture. The Langevin capture means that the spiral trajectory results in "closer interaction" and formation of the ion-molecular complex, which then can either spiral out or provide inelastic changes of state and formation of different secondary products. This ion-molecular capture process based on polarization occurs when the interaction energy between charge and induced dipole $P_m E = \alpha\frac{e}{4\pi r^2}\frac{e}{4\pi\varepsilon_0 r^2}$ becomes the order of the kinetic energy of the colliding partners $\frac{1}{2}Mv^2$, where M is their reduced mass, and v their relative velocity. From the equality of kinetic and interaction energies, the **Langevin cross section** of the polarization capture can be found as $\sigma_L \approx \pi r^2$:

$$\sigma_L = \sqrt{\frac{\pi\alpha e^2}{\varepsilon_0 M v^2}}. \tag{2.82}$$

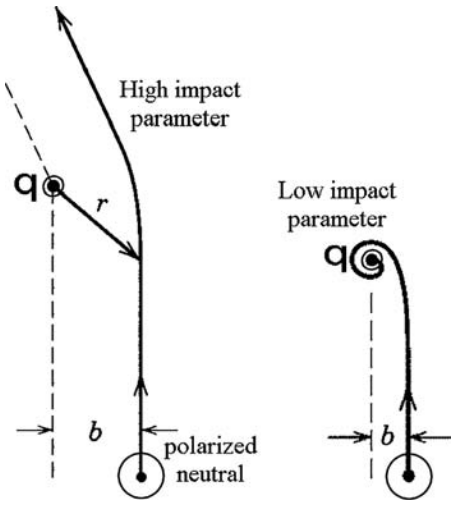



**FIGURE 2.12** The Langevin scattering in the polarization potential.

Detailed derivation of the exact formula for Langevin corss section can be found in the book of M.A. Lieberman and A.J. Lichtenberg (1994). The Langevin capture rate coefficient $k_L \approx \sigma_{L\nu}$ according to Eq. (2.82) does not depend on velocity, and therefore does not depend on temperature:

$$k_L = \sqrt{\frac{\pi\alpha e^2}{\varepsilon_0 \mathrm{M}}} \tag{2.83}$$

It is convenient for calculations to express the polarizability $\alpha$ in $10^{-24}$ cm$^3$ (cubic Angstroms, see Table 2.12), reduced mass in atomic mass units (amu), and then to rewrite and use Eq. (2.83) in the following numerical form:

$$k_L^{ion/neutral} = 2.3\times10^{-9}\ \mathrm{cm}^3/\mathrm{sec}^* \sqrt{\frac{\alpha,10^{-24}\mathrm{cm}^3}{\mathrm{M,amu}}} \tag{2.84}$$

The typical value of the Langevin rate coefficient for ion-molecular reactions is $10^{-9}$ cm$^3$/s, which is 10 times higher than the typical value of gas-kinetic rate coefficient for binary collisions of neutral particles: $k_0 \approx 10^{-10}$ cm$^3$/s. Also, $\alpha$/M in Eq. (2.84) reflects the "specific volume" of an atom or molecule, because the polarizability $\alpha$ actually corresponds to the particle volume. This means that the Langevin capture rate coefficient grows with a decrease of "density" of an atom or molecule. The above relations described interactions of a charge with an induced dipole. If an ion interacts with a molecule having not only an induced dipole but also a permanent dipole moment $\mu_{D,}$ then the Langivin capture cross section becomes larger. The corresponding correction of Eq. (2.83) was done by Su and Bowers (1973):

$$k_L = \sqrt{\frac{\pi e^2}{\varepsilon_0 \mathrm{M}}}\left(\sqrt{\alpha} + c\mu_D\sqrt{\frac{2}{\pi T_0}}\right) \tag{2.85}$$

Here $T_0$ is a gas temperature (in energy units) and parameter $0 < c < 1$ describes the effectiveness of a dipole orientation in the electric field of an ion. For molecules like $H_20$ or HF, having large a permanent dipole moment, the ratio of the second to the first terms in Eq. (2.85) is about $\sqrt{\mathrm{I}/\mathrm{T}_0}$, where I is ionization potential. It means that the Langevin cross sections and rate coefficients for dipole molecules and radicals can exceed by a factor of 10 the numerical values obtained for pure polarization collisions Eq. (2.84). Derivation of the Langevin formula Eq. (2.82) did not specify the mass of a charged particle. This means that the Langevin cross section and rate coefficients Eqs. (2.82) and (2.83) also can be applied for an electron–neutral interaction. In this case, the reduced mass M is close to an electron mass and numerical formula for the Langevin rate coefficient for the electron–neutral collision can be presented as:

$$\kappa_L^{electron/neutral} = 10^{-7}\ \mathrm{cm}^3/\mathrm{s}^* \sqrt{\alpha,10^{-24}}\mathrm{cm}^3 \tag{2.86}$$

Numerical value of the Langevin rate coefficient for the electron–neutral collision is about $10^{-7}$ cm$^3$/s and does not depend on temperature.

### 2.6.2 The Ion-Atom Charge Transfer Processes

An electron can be transferred during a collision from a neutral particle to a positive ion or from a negative ion to a neutral particle. These processes are referred to as charge transfer or charge exchange. The charge exchange reaction without significant defect of the electronic state energy $\Delta$E during collision is called the **resonant charge transfer**. Otherwise, charge transfer is called **non-resonant**. The resonant charge transfer is a non-adiabatic process and usually has a very large cross

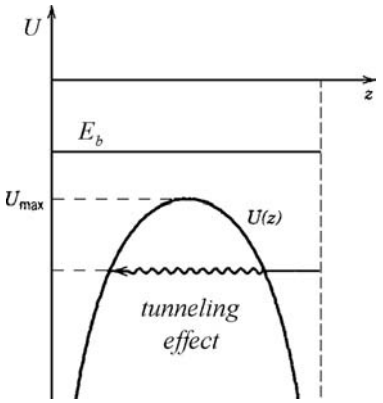



**FIGURE 2.13** Energy terms for the resonant charge exchange.

section. Consider a charge exchange between a neutral particle B and a positive ion $A^+$ assuming the particle $B/B^+$ being at rest:

$$A^+ + B \rightarrow A + B^+ \tag{2.87}$$

The energy scheme of this reaction is illustrated in Figure 2.13, where one can see the Coulomb potential energy of an electron in the Coulomb field of $A^+$ and $B^+$:

$$U(z) = -\frac{e^2}{4\pi\varepsilon_0 z} - \frac{e^2}{4\pi\varepsilon_0 |r_{AB} - z|} \tag{2.88}$$

Here $r_{AB}$ is a distance between the centers of A and B. The maximum value of the potential energy corresponds to $z = r_{AB}/2$ and is equal to:

$$U_{\max} = -\frac{e^2}{\pi\varepsilon_0 r_{AB}} \tag{2.89}$$

The charge transfer, Eq. (2.87), is possible in the framework of classical mechanics (see Figure 2.13) if the maximum of potential energy $U_{max}$ is lower than the initial energy $E_B$ of an electron which is going to be transferred from level n of particle B:

$$E_B = -\frac{I_B}{n^2} - \frac{e^2}{4\pi\varepsilon_0 r_{AB}} \geq U_{\max} \tag{2.90}$$

Here $I_{AB}$ is the ionization potential of atom B. From Eqs. (2.89) and (2.90), the maximum distance between the interacting heavy particles is found when the charge transfer is still permitted by classical mechanics:

$$r_{AB}^{\max} = \frac{3e^2 n^2}{4\pi\varepsilon_0 I_B} \tag{2.91}$$

If the charge exchange is resonant and therefore not limited by the defect of energy, the classical reaction cross section can be found from Eq. (2.91) as $\pi r_{AB}^2$:

$$\sigma_{chtr}^{class} = \frac{9e^4 n^4}{16\pi\varepsilon_0^2 I_B^2}. \tag{2.92}$$

The classical cross section of charge exchange does not depend on the kinetic energy of the interacting species and its numerical value for ground state transfer (n = 1) is approximately the gas-kinetic cross section for collisions of neutrals. The actual cross section of a resonant charge transfer can be much higher than Eq. (2.92), taking into account the quantum mechanical effect of electron tunneling from B to $A^+$ (see Figure 2.13). This effect can be estimated by calculating the electron tunneling probability $P_{tunn}$ across a potential barrier of height about $I_B$ and width d:

$$P_{tunn} \approx \exp\left(-\frac{2d}{\hbar}\sqrt{2meI_B}\right). \tag{2.93}$$

The frequency of electron oscillation in the ground state atom B is about $I_B/\hbar$, so the frequency of the tunneling is $I_B P_{tunn}/\hbar$. Taking into account Eq. (2.93), the maximum barrier width $d_{max}$, when the tunneling frequency still exceeds the reverse ion-neutral collision time: $I_B P_{tunn}/\hbar > v/d$ and as a result, the tunneling can take place. Here v is the relative velocity of the colliding partners. Then the cross section $\pi d_{\max}^2$ of the electron tunneling from B to $A^+$, leading to the resonance charge exchange is:

$$\sigma_{ch.tr}^{tunn} \approx \frac{1}{I_B}\left(\frac{\pi\hbar^2}{8me}\right)\left(\ln\frac{I_B d}{\hbar} - \ln v\right)^2. \tag{2.94}$$

More detailed derivation of the electron tunneling cross section (Rapp, Francis, 1962) results in more complicated expressions, which can be presented in a numerical form very similar to Eq. (2.94):

$$\sqrt{\sigma_{ch.tr}^{tunn}\text{cm}^2} \approx \frac{1}{\sqrt{I_B\text{eV}}}\left(6.5*10^{-7} - 3*10^{-8}\ln v, \frac{\text{cm}}{\text{s}}\right). \tag{2.95}$$

This numerical formula can be applied to calculate the resonant charge transfer cross section in the velocity range v = $10^5$–$10^8$ cm/s. At higher energies, when the velocity v exceeds $10^8$ cm/s, the tunneling cross section decreases to the gas-kinetic cross sections (about $3*10^{16}$ $cm^2$) and then it is stabilized. The stabilized cross section does not depend on energy and is given by the classical expression Eq. (2.92).

At the lower limit of application of Eq. (2.95), v = $10^5$ cm/s and the tunneling cross section reaches values as high as $10^{-14}$ $cm^2$. Equations (2.94) and (2.95) were derived assuming straight line collision trajectories. Therefore applying Eqs. (2.95) and (2.95) at lower velocities is impossible because when $v \le 10^5 \frac{\text{cm}}{\text{s}} / \sqrt{\text{M}}$ (M is a reduced mass), the collision trajectory is strongly perturbed during the charge exchange. In low energy collision, when $v \le 10^5 \frac{\text{cm}}{\text{s}} / \sqrt{\text{M}}$, the ion and neutral are captured in a complex. In this case, the exchanging electron always has enough time for tunneling and therefore the equal probability ½ are found on either particle. Then the resonant charge exchange cross section can be found as half of the Langevin capture cross section Eq. (2.82):

$$\sigma_L = \frac{1}{2}\sqrt{\frac{\pi\alpha e^2}{\varepsilon_0 \text{M}}}\frac{1}{v}. \tag{2.96}$$

The typical numerical value of this cross section at room temperature is about $10^{-14}$ $cm^2$ and higher. It grows as the velocity (and temperature) decreases even faster than in the case of intermediate velocities Eq. (2.95).

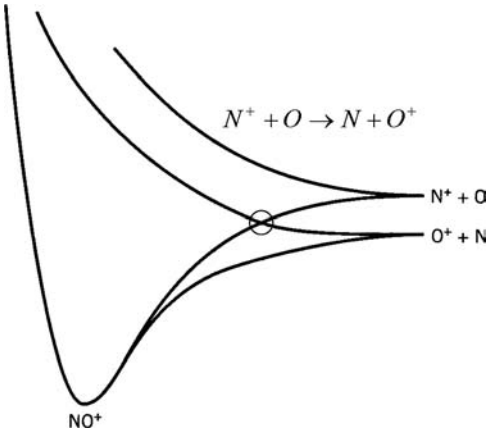



**FIGURE 2.14** Terms illustrating the non-resonant charge exchange.

### 2.6.3 The Non-Resonant Charge Transfer Processes

Consider an electron transfer between oxygen and nitrogen atoms as an example of the non-resonant charge exchange:

$$N^+ + O \rightarrow N + O^+, \quad \Delta E = -0.9 \text{ eV}. \tag{2.97}$$

The principal potential curves, illustrating the nonresonant charge transfer Eq. (2.97) are shown in Figure 2.14. From Table 2.1, the ionization potential of oxygen (I = 13.6 eV) is lower than that of nitrogen (I = 14.5 eV). This is the reason that the electron transfer from oxygen to nitrogen is an exothermic process and the separated $N + O^+$ energy level is located 0.9 eV lower than the separated $O + N^+$ energy level. The reaction (2.97) begins with $N^+$ approaching O by following the attractive $NO^+$ term. Then this term is crossing with the repulsive $NO^+$ term and the system experiences a non-adiabatic transfer, which results in the formation of $O^+ + N$. The excess energy of approximately 1 eV goes into the kinetic energy of the products. The cross section of such exothermic charge exchange reactions at low thermal energies is of the order of the resonant cross sections of tunneling Eq. (2.95) or Langevin capture Eq. (2.96). The endothermic reactions of charge exchange, like the reverse process $N + O^+ \rightarrow N^+ + O$, are usually very slow at low gas temperatures with low energies of the colliding ions and heavy neutral particles.

The important role of the non-resonant charge exchange processes can be illustrated by the so-called effect of **acidic behavior of nonthermal air plasma**. Ionization of air in nonthermal discharges primarily leads to a large amount of $N_2^+$ ions (with respect to other positive ions) because of the high molar fraction of nitrogen in air. Later, low ionization potential and high dipole moment of water molecules can result in fast charge exchange:

$$N_2^+ + H_2O \rightarrow N_2 + H_2O^+, \quad k(300K) = 2.2 * 10^{-9} \text{cm}^3 / \text{s}. \tag{2.98}$$

The whole ionization process can be significantly focused on the formation of water ions $H_2O^+$ even though the molar fraction of water in air is low. The generated water ions can then react with neutral water molecules in the quite fast ion-molecular reaction:

$$H_2O^+ + H_2O \rightarrow H_3O + OH, \Delta H = -12 \text{ kcal / mole}, \text{k}(350\text{K}) = 0.5 * 10^{-9} \text{ cm}^3 / \text{s}.a \tag{2.99}$$

As a result, the production of $H_3O^+$-ions and OH-radicals can be observed which provides the acidic behavior of the air plasma. The selective generation of OH-radicals in nonthermal air discharges is the fundamental basis for employing discharges for purifying air from different pollutants.

### 2.6.4 The Ion-Molecular Reactions with Rearrangement of Chemical Bonds

Reaction Eq. (2.99) demonstrates the acidic behavior of nonthermal air plasma. This class of ion-molecular processes consists of very different reactions, which can be subdivided into many groups, including:

- $(A)B^+ + C \rightarrow A + (C)B^+$, reactions with a positive ion transfer,
- $A(B^+) + C \rightarrow (B^+) + AC$, reactions with a neutral transfer,
- $A(B^+) + (C)D \rightarrow (A)D + (C)B^+$, double exchange reactions,
- $A(B^+) + (C)D \rightarrow AC^+ + BD$, reconstruction processes.

These groups of processes are shown above with positive ions, but similar reactions take place with negative ions. The special feature of the ion-molecular reactions making them very distinctive from regular atom-molecular processes between neutral species can be stated as the following statement: ***Most Exothermic Ion-Molecular Reactions Have No Activation Energy*** (V.L. Talrose, 1952). This is because quantum mechanical repulsion between molecules, which provides the activation barrier even in the exothermic reactions of neutrals, can be suppressed by the charge-dipole attraction in ion-molecular reactions. Thus, the rate coefficients are very large and can be found based on the Langevin relations (2.83) and (2.85). Not every Langevin capture leads to the ion-molecular reactions with complicated rearrangement of chemical bonds or to the orbital symmetry and spin-forbidden reactions. For this reason, the pre-exponential Arrhenius factors of the complicated exchange reaction, orbital symmetry, and spin forbidden reactions can be lower than the Langevin rate coefficients (T. Su, M.T. Bowers, 1995). In general, however, the exothermic ion-molecular reactions with rearrangement of chemical bonds are much faster than the corresponding processes between neutrals. This effect is especially important at low temperatures when even small activation barriers can dramatically slow down exothermic reactions. An extensive listing of the ion-molecular reaction rate coefficients is presented in the monograph of L. Virin, R. Dgagaspanian, G. Karachevtsev, V. Potapov, and V. Talrose (1978).

### 2.6.5 Ion-Molecular Chain Reactions and Plasma Catalysis

The absence of activation energies in exothermic ion-molecular reactions facilitates the organization of chain reactions in ionized media. Thus, for example, the important industrial process of $SO_2$ oxidation into $SO_3$ and sulfuric acid is highly exothermic; however, it can not be arranged as an effective chain process without a catalyst, because of the energy barriers of intermediate elementary reactions between neutral species. However, the long length $SO_2$ chain oxidation becomes effective in ionized media through the negative ion mechanism in water or in droplets in heterogeneous plasma (Daniel, Jacob, 1986; B. Potapkin, M. Deminsky, A. Fridman, V. Rusanov, 1995).

This chain oxidation process begins with the $SO_2$ dissolving and forming a negative ion $SO_3^{2-}$ in the case of low acidity (pH > 6.5). Then, a charge transfer process with a OH-radical results in forming an active $SO_3^-$ ion-radical and subsequently initiating a chain reaction:

$$OH + SO_3^{2-} \rightarrow OH^- + SO_3^- \tag{2.100}$$

The active $SO_3^-$ ion-radical becomes the main chain-carrying particle, initiating the chain mechanism. The first reaction of chain propagation is the attachment of an oxygen molecule:

$$SO_3^- + O_2 \rightarrow SO_5^-, \quad k = 2.5 * 10^{-12} \text{cm}^3 / \text{s}. \tag{2.101}$$

Then the chain propagation can go either through the intermediate formation of $SO_4^-$ ion –radical:

$$SO_5^- + SO_3^{2-} \rightarrow SO_4^{2-} + SO_4^-, \quad k = 5 * 10^{-14} \text{cm}^3 / \text{s}. \tag{2.102}$$

$$SO_4^- + SO_3^{2-} \rightarrow SO_4^{2-} + SO_3^-, \quad k = 3.3 * 10^{-12} \text{cm}^3 / \text{s} \tag{2.103}$$

or similarly through the intermediate formation of $SO_5^{2-}$ ion- radical:

$$SO_5^- + SO_3^{2-} \rightarrow SO_5^{2-} + SO_3^-, \quad k = 1.7 * 10^{-14} \text{cm}^3 / \text{s} \tag{2.104}$$

$$SO_5^{2-} + SO_3^{2-} \rightarrow SO_4^{2-} + SO_4^{2-}, \quad k = 2 * 10^{-14} \text{cm}^3 / \text{s} \tag{2.105}$$

Considering all these reactions, one should take into account that the ion $SO_4^{2-}$ is a product of the $SO_2$ oxidation. This stable ion corresponds to the $SO_3^-$ molecule in the gas phase and to sulfuric acid $H_2S$ in a water solution. Here, the chain termination is due to losses of $SO_3^-$ and $SO_5^-$, and the chain length easily exceeds $10^3$. Thus it can be concluded that plasma as an ionized medium permits avoiding activation energy barriers of elementary exothermic reactions and as a result can operate similarly to traditional catalysts. This phenomenon is called **plasma catalysis** (B. Potapkin, V. Rusanov, S.A. Fridman, 1989).

### 2.6.6 Ion-Molecular Processes of Cluster Growth, the Winchester Mechanism

From Section 2.6.1, it is clear that ion-molecular reactions are very favorable to clusterization due to the effective long-distance charge-dipole and polarization interaction leading to Langevin capturing. Indeed both positive and negative ion-molecular reactions can make a fundamental contribution in the nucleation and the cluster growth phases of dusty plasma generation in electric discharges (A. Bouchoule, 1993; A. Bouchoule, 1999) and even soot formation in combustion. Specific details of cluster growth in nonthermal discharges will be discussed during consideration of the physics and chemistry of the dusty plasmas. Here the fundamental effects of ion-molecular processes of cluster growth will be pointed out. Besides the effect of Langevin capture (Section 2.6.1), the less known, but very important **Winchester Mechanism** of ion-molecular cluster growth needs to be considered. (L. Boougaenko, L. Polak, 1989). A key point of the Winchester Mechanism is the thermodynamic advantage of the ion-cluster growth processes. If a sequence of negative ions defining the cluster growth is considered:

$$A_1^- \rightarrow A_2^- \rightarrow A_3^- \rightarrow \ldots \rightarrow A_n^- \rightarrow \ldots \tag{2.106}$$

the corresponding electron affinities $EA_1$, $EA_2$, $EA_3$, …, $EA_n$, … are usually increasing, ultimately reaching the value of the work function (electron extraction energy), which is generally larger than the EA for small molecules. As a result, each elementary step reaction of the cluster growth $A_n^- \rightarrow A_{n+1}^-$ has an apriori tendency to be exothermic. Exothermic ion-molecular reactions have no activation barrier and are usually very fast, thus the Winchester Mechanism explains effective cluster growth based on ion-molecular processes. The phenomenon is important until

the cluster becomes too large and the difference in electron affinities $EA^-_{n+1} - EA_n$ becomes negligible. Each elementary step $A^-_n \rightarrow A^-_{n+1}$ includes the cluster rearrangements with an electron usually going to the furthest end of the complex. This explains the origin of the term "Winchester", which sticks to the described mechanism of cluster growth. The Winchester Mechanism is valid not only for negative but also for positive ion clusters. The thermodynamic advantage of ion-cluster growth for positive ion also holds. If a sequence of positive ions defining the cluster growth is considered:

$$A^+_1 \rightarrow A^+_2 \rightarrow A^+_3 \rightarrow \ldots \rightarrow A^+_n \rightarrow \ldots \tag{2.107}$$

the corresponding ionization energies $I(A_1)$, $I(A_2)$, $I(A_3)$, …, $I(A_n)$, … are usually decreasing to ultimately reach the value of the work function (electron extraction energy), which is generally lower than ionization energy for small molecules. As a result, each elementary step reaction of the cluster growth $A^+_n \rightarrow A^+_{n+1}$ has an a priori tendency to be exothermic for positive ions as well. Thus, the Winchester Mechanism explains effective cluster growth for both positive and negative ions. As a specific example, one can mention a sequence of ion-molecular reactions during the nucleation stage and cluster growth (dusty plasma formation) in low-pressure silane $SiH_4$ and silane-argon $SiH_4$ – Ar discharges (A. Garscadden, 1994; A. A. Howling e.a., 1993a; A. Fridman, L. Boufendi, A. Bouchoule, e.a., 1996a):

$$Si_nH^-_{2n+1} + SiH_4 \rightarrow Si_{n+1}H^-_{2n+3} + H_2 \tag{2.108}$$

The nucleation process can be initiated by a dissociative attachment: $e + SiH_4 + SiH_3^- + H$ (k = $10^{-12}$ cm$^3$/s at electron temperature $T_e$ = 2 eV) and then continues by the sequence (2.108) of exothermic, thermo-neutral, and slightly endothermic reactions as it is illustrated in Figure 2.15. Thermal effects of the first 4 reactions of Eq. (2.108) demonstrate the Winchester mechanism in silane plasma:

$$SiH_3^- + SiH_4 \rightarrow Si_2H_5^- + H_2 - 0.07eV \tag{2.109}$$

$$Si_2H_5^- + SiH_4 \rightarrow Si_3H_7^- + H_2 + 0.07eV \tag{2.110}$$

$$Si_3H_7^- + SiH_4 \rightarrow Si_4H_9^- + H_2 + 0.07eV \tag{2.111}$$

$$Si_4H_9^- + SiH_4 \rightarrow Si_5H_{11}^- + H_2 + 0.00eV \tag{2.112}$$

The Winchester mechanism shows the tendency of energy effects on ion-cluster growth. Not all reactions of the sequence are exothermic, thermo-neutral and even endothermic stages can be found. This results in "bottle neck" phenomena in nucleation kinetics in low pressure silane discharges.

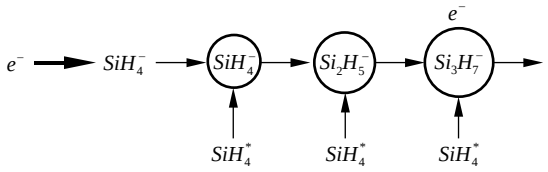



**FIGURE 2.15** Illustration of ion-molecular reactions of cluster growth.

## 2.7 PROBLEMS AND CONCEPT QUESTIONS

### 2.7.1 Electron Energy Distribution Functions

If the electron energy distribution function is Maxwellian and the electron temperature is $T_e$ = 1 eV, (1) what is the mean velocity <v> of the electrons in this case? (2) what is the mean electron energy?, (3) which electron energy $m\langle v\rangle^2/2$ corresponds to the mean velocity? (4) how can you explain the difference between the two "average electron energies" in terms of standard deviation?

### 2.7.2 Ionization Potentials and Electron Affinities

Why are most ionization potentials (Table 2.1) greater than electron affinities (Table 2.2)?

### 2.7.3 Positive and Negative Ions

Why is it very doable to produce the double- or multi-charged positive ions such as $A^{++}$ or $A^{+++}$ in plasma, and impossible to generate in a practical gas discharge plasmas the double or multi-charged negative ions such as $A^-$ or $A^{---}$?

### 2.7.4 Mean Free Path of Electrons

In which gases are the mean free path of electrons with electron temperature $T_e$ = 1eV longer and why? (helium, nitrogen or water vapor at similar pressure conditions)?

### 2.7.5 Reaction Rate Coefficients

Recalculate the Maxwellian electron energy distribution function into the electron velocity distribution function f(v) and then find an expression for reaction rate coefficient Eq. (2.7) in the case when $\sigma = \sigma_0 = const$.

### 2.7.6 Elastic Scattering

Charged particle scattering on neutral molecules with permanent dipole momentum can be characterized by the following cross section dependence on energy: $\sigma(\varepsilon) = Const/\varepsilon$. How does the rate coefficient of the scattering depend on temperature in this case?

### 2.7.7 Direct Ionization by Electron Impact

Using Eq. (2.18) for the general ionization function f(x), find the electron energy corresponding to the maximum value of direct ionization cross section. Compare this energy with the electron energy optimal for direct ionization according to the Thomson formula.

### 2.7.8 Comparison of Direct and Stepwise Ionization

Why does the direct ionization make a dominant contribution in nonthermal electric discharges while in thermal plasmas stepwise ionization is more important?

### 2.7.9 Stepwise Ionization

Estimate a stepwise ionization reaction rate in Ar at electron temperature 1 eV, assuming quasi-equilibrium between plasma electrons and electronic excitation of atoms, and using Eq. (2.29).

### 2.7.10 Electron Beam Propagation in Gases

How does the propagation length of a high energy electron beam depend on pressure at fixed temperature? How does it depend on the temperature at a fixed pressure?

### 2.7.11 Photoionization

Why can the photoionization effect play the dominant role in the propagation of both nonthermal and thermal discharges in fast flows including supersonic ones? What is the contribution of photoionization in the propagation of slow discharges?

### 2.7.12 Massey Parameter

Estimate the adiabatic Massey parameter for ionization of Ar-atom at rest in collision with $Ar^+$-ion, having kinetic energy twice exceeding the ionization potential. Discuss the probability of ionization in this case.

### 2.7.13 Ionization in an Ion-Neutral Collision

How large is an $H^+$-atom energy supposed to be for this ion to reach a velocity the same as an electron velocity in a hydrogen atom? Compare this energy with the H-atom ionization potential. What is your conclusion about the ionization process initiated by ion impact?

### 2.7.14 Ionization in Collision of Excited Heavy Particles

Why is ionization in the collision of vibrationally excited molecules usually much less effective than in a similar collision of electronically excited molecules.

### 2.7.15 Ionization in Collision of Vibrationally Excited Molecules

In principle, ionization can take place in collision of nitrogen molecules having 32 vibrational quanta each. Is it possible to increase the probability of such ionization by increasing the vibrational energy of the collision partners or by increasing the electronic energy of the collision partners?

### 2.7.16 Dissociative Electron–Ion Recombination

According to Eq. (2.44), a rate coefficient of dissociative recombination is reversibly proportional to gas temperature. However in reality, when the electric field is fixed, this rate coefficient is decreasing faster than the reverse proportionality. How can you explain that?

### 2.7.17 Ion-Conversion, Preceding Electron–Ion Recombination

How large is the relative concentration of complex ions like $N_2^+\left(N_2\right)$ supposed to be to compete with conventional electron recombination with $N_2^+$-ions in the same discharge conditions?

### 2.7.18 Three-Body Electron–Ion Recombination

Derive the reaction rate coefficient of the three-body recombination, using Eq. (2.26) for the stepwise ionization rate coefficient and the Saha thermodynamic equation for ionization/recombination balance.

### 2.7.19 Elementary Processes of Charged Species in Plasma

Give an example of discharge parameters (type of gas, range of pressures, plasma densities, etc.), when the radiative electron–ion recombination dominates between other mechanisms of losses of charged particles.

### 2.7.20 Dissociative Attachment

Calculate the dissociative attachment rate coefficients for molecular hydrogen and molecular oxygen at electron temperatures 1 eV and 5 eV. Compare the results and give your comments.

### 2.7.21 Negative Ions in Oxygen

What are the discharge and gas parameters supposed to be if the negative oxygen ions are mostly generated by three-body attachment processes, and not by dissociative attachment?

### 2.7.22 Associative Detachment

Calculate the relative concentration of CO in $CO_2$-discharge plasma if the main associative detachment process in this discharge dominates over dissociative attachment. Suppose that the electron temperature is 1 eV, and the concentration of negative ions is 10 times less than positive ions.

### 2.7.23 Detachment by Electron Impact

Detachment of an electron from a negative ion by electron impact can be considered as "ionization" of the negative ion by electron impact. Why can't the Thomson formula be used in this case instead of Eq. (2.64)?

### 2.7.24 Negative Ions in Thermal Plasma

Why is the concentration of negative ions in thermal plasmas usually very low even in electro-negative gases? Which specific features of thermal plasmas with respect to nonthermal plasmas are responsible for that?

### 2.7.25 Ion–Ion Recombination in Binary Collisions

Determine how the ion–ion recombination rate coefficient depends on temperature. Use Eq. (2.70) for the cross section of the binary collision recombination rate and assume the Maxwellian distribution function with temperature $T_0$ for both positive and negative ions.

### 2.7.26 Three-Body Ion–Ion Recombination

Comparing Thomson's and Langevin's approaches for the three-body ion–ion recombination and correspondent rate coefficients Eqs. (2.74) and (2.62), find a typical value of pressure when the recombination reaches the maximum rate. Consider oxygen plasma with heavy particle temperature 1 eV as an example.

### 2.7.27 Langevin Cross Section

Calculate the Langevin cross section for $Ar^+$-ion collision with an argon atom and with a water molecule. Comment on your result. Compare the contribution of the polarization term and the charge-dipole interaction term in the case of $Ar^+$-ion collision with a water molecule.

### 2.7.28 Resonant Charge Transfer Process

Is it possible to observe the resonant charge transfer process during an ion-neutral interaction with only a slight perturbation of trajectories of the collision partners?

### 2.7.29 Tunneling Effect in Charge Transfer

Estimate how strong the tunneling effect can be in resonant charge transfer in ion-neutral collision. For two different velocities compare the cross section of charge exchange between an argon atom and argon positive ion calculated with and without taking into account the tunneling effect.

### 2.7.30 Non-Resonant Charge Exchange

Most negative ions in nonthermal atmospheric pressure air discharges are in form of $O_2^-$. What will happen to the negative ion distribution if even a very small amount of fluorine compounds is added to air?

### 2.7.31 Plasma Catalytic Effect

Compare the catalytic effect of plasma provided by negative ions, positive ions, active atoms, and radicals. All these species are usually quite active, but which one is the most generally relevant to the plasma catalysis.

### 2.7.32 Winchester Mechanism

Why is the ion mechanism of cluster growth usually less effective in combustion plasma during soot formation than in similar situations in electrical discharge plasma?

# 3 Elementary Processes of Excited Molecules and Atoms in Plasma

## 3.1 ELECTRONICALLY EXCITED ATOMS AND MOLECULES IN PLASMA

The extremely high chemical activity of plasma is based on high and quite often a super-equilibrium concentration of active species. The active species generated in plasma include chemically aggressive atoms and radicals, charged particles – electron and ions, and excited atoms and molecules. Elementary processes of atoms and radicals are traditionally considered in a framework of chemical kinetics (H. Eyring, S.H. Lin, S.M. Lin, 1980; V. Kondratiev, E. Nikitin, 1981), elementary processes of the charged particles were considered in Chapter 2, elementary processes of the excited species are considered in this chapter. Excited species, in particular vibrationally excited molecules, are of special importance in plasma chemical kinetics because most of the discharge energy in molecular gases (often more than 95%) can be focused on vibrational excitations by electron impact. Excited species can be subdivided into three groups: electronically excited atoms and molecules, vibrationally excited molecules, and rotationally excited molecules.

### 3.1.1 Electronically Excited Particles, Resonance, and Metastable States

High electron energies in electric discharges provide a high excitation rate of different electronically excited states of atoms and molecules by electron impact. The energy of the electronically excited particles usually is relatively high (about 5–10 eV), but their lifetime is generally very short (usually about $10^{-8}$–$10^{-6}$ s). If a radiative transition to the ground state is not forbidden by quantum mechanical selection rules, such a state is called the **resonance excited state.** The resonance states have the shortest lifetime (about $10^{-8}$ s) with respect to radiation, and therefore their direct contribution in the kinetics of chemical reactions in plasma is usually small. If the radiative transition is forbidden by selection rules, the lifetime of the excited particles can be much longer because of the absence of spontaneous transition. Such states are called the **metastable excited states**. The energy of the first resonance excited states as well as the energy of the low energy metastable states and their radiative lifetime are given in Table 3.1. Because of their long life with respect to radiation, the metastable electronically excited atoms and molecules are able to accumulate the necessary discharge energy and significantly contribute to the kinetics of different chemical reactions in plasma. The metastable excited particles can lose their energy not only by radiation but also by means of different collisional relaxation processes. The energy of excitation of the metastable states can be quite low (sometimes less than 1 eV) and therefore their concentration in electric discharges can be high.

### 3.1.2 Electronically Excited Atoms

The collision of a high-energy plasma electron with a neutral atom in a ground state can result in energy transfer from the free plasma electron to a bound electron in the atom. The most important

**TABLE 3.1**
**The Lowest Electronically Excited States and the Lowest Metastable States for Excited Atoms with Their Radiative Lifetimes**

| Atom and it's Ground State | First Resonance Excited States | Resonance Energy | Low Energy Metastable States | Metastable's Energy | Metastable's Lifetime |
|---|---|---|---|---|---|
| He ($1s^2\ ^1S_0$) | $2p\ ^1P_1^0$ | 21.2 eV | $2s\ ^3S_1$ | 19.8 eV | $2*10^{-2}$ s |
| He($1s^2\ ^1S_0$) | | | $2s\ ^1S_0$ | 20.6 eV | $9*10^3$ s |
| Ne ($2s^2p^6\ ^1S_0$) | $3s\ ^1P_1^0$ | 16.8 eV | $4s\ ^3P_2$ | 16.6 eV | $4*10^2$ s |
| Ne ($2s^2p^6\ ^1S_0$) | | | $4s\ ^3P_0$ | 16.7 eV | 20 s |
| Ar ($3s^2p^6\ ^1S_0$) | $4s\ ^2P_1^0$ | 11.6 eV | $4s\ ^2P_{0,2}^0$ | 11.6 eV | 40 s |
| Kr ($4s^2p^6\ ^1S_0$) | $5s\ ^3P_1^0$ | 10.0 eV | $5s\ ^3P_2$ | 9.9 eV | 2 s |
| Kr ($4s^2p^6\ ^1S_0$) | | | $5s\ ^3P_0$ | 10.6 eV | 1 s |
| H ($1s\ ^2S_{1/2}$) | $2p\ ^2P_{1/2,3/2}^0$ | 10.2 eV | $2s\ ^2S_{1/2}$ | 10.2 eV | 0.1 s |
| N ($2s^2p^3\ ^4S_{3/2}$) | $3s\ ^4P_{1/2,3/2,5/2}^0$ | 10.3 eV | $2p^3\ ^2D_{3/2}$ | 2.4 eV | $6*10^4$ s |
| N ($2s^2p^3\ ^4S_{3/2}$) | | | $2p^3\ ^2D_{5/2}$ | 2.4 eV | $1.4*10^5$ s |
| N ($2s^2p^3\ ^4S_{3/2}$) | | | $2p^3\ ^2P_{1/2}^0$ | 3.6 eV | 40 s |
| N ($2s^2p^3\ ^4S_{3/2}$) | | | $2p^3\ ^2P_{3/2}^0$ | 3.6 eV | $1.7*10^2$ s |
| O ($2s^2p^4\ ^3P_2$) | $3s\ ^3S_1^0$ | 9.5 eV | $2p^4\ ^3P_1$ | 0.02 eV | – |
| O ($2s^2p^4\ ^3P_2$) | | | $2p^4\ ^3P_0$ | 0.03 eV | – |
| O ($2s^2p^4\ ^3P_2$) | | | $2p^4\ ^1D_2$ | 2.0 eV | $10^2$ s |
| O ($2s^2p^4\ ^3P_2$) | | | $2p^4\ ^1S_0$ | 4.2 eV | 1 s |
| Cl ($3s^2p^5\ ^2P_{3/2}$) | $4s\ ^2P_{1/2,3/2}^0$ | 9.2 eV | $3p^5\ ^3P_{1/2}^0$ | 0.1 eV | $10^2$ s |

growth of energy of a bound electron during the excitation is due to an increase in the **principal quantum number "n"**, but it also usually grows with the value of **angular momentum quantum number "l"**. This effect can be illustrated quantitatively in a case of strong excitation of any atom to a relatively high principal quantum number "n". In this case, one electron is moving quite far from the nucleus and therefore it is almost moving in a Coulomb potential. Such an excited atom is somewhat similar to a hydrogen atom. Its energy can be described by the **Bohr formula with the Rydberg correction** term $\Delta_l$, which takes into account the short-term deviation from the Coulomb potential when the electron approaches the nucleus (see L. Landau, 1997):

$$E = -\frac{me^4}{2\hbar^2\left(4\pi\varepsilon_0\right)^2}\frac{1}{\left(n+\Delta_l\right)^2}. \tag{3.1}$$

The Rydberg correction term $\Delta_l = 0$ in hydrogen atoms, and Eq. (3.1) becomes identical to the conventional Bohr formula. In general, $\Delta_l$ does not depend on the principal quantum number "n", but depends on the angular momentum quantum number "l" of the excited electron as well as on the L and S momentum of the whole atom. If the L and S quantum numbers are fixed, the Rydberg correction term $\Delta_l$ decreases fast with increasing angular momentum quantum number "l". When the quantum number "l" is growing, the excited electron spends less time in the vicinity of the nucleus and the energy levels are closer to those of hydrogen. This effect can be demonstrated by numerical values of $\Delta_l$ for excited helium both in case of S = 0 and S = 1 (see Table 3.2). In general, energy levels of excited atoms depend not only on the principal quantum number "n" of the excited electrons and the **total angular orbital momentum "L"** but also on the **total spin number "S"** and **total momentum quantum number "J"**. Thus, typically, the triplet terms (S = 1) lie below the correspondent singlet terms (S = 0) and hence the energy levels in corresponding excited, atomic states are lower for the higher total spin numbers.